

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280186

Kind Code

A1

Publication Date

September 04, 2025

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SUBSTRATE FOR IMAGE SENSOR

Abstract

A substrate for an image sensor according to an embodiment includes an insulating layer; and a conductive pattern portion disposed on the insulating layer, wherein the insulating layer comprises: a first insulating part; and a second insulating part surrounding the first insulating part and spaced apart from the first insulating part with an open region therebetween, wherein the conductive pattern portion comprises a first lead pattern part disposed on the first insulating part; a second lead pattern part disposed on the second insulating part; and an extension pattern part disposed on the open region of the insulating layer to connect the first lead pattern part with the second lead pattern part and including a bent portion.

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Family ID: 70773146

Appl. No.: 19/211982

Filed: May 19, 2025

Foreign Application Priority Data

KR 10-2018-0143392

Nov. 20, 2018

KR 10-2019-0002156

Jan. 08, 2019

Related U.S. Application Data

parent US continuation 18176140 20230228 parent-grant-document US 12335589 child US 19211982

parent US continuation 17294625 20210517 parent-grant-document US 11622064 WO

continuation PCT/KR2019/015531 20191114 child US 18176140

Publication Classification

Int. Cl.: **H04N23/54** (20230101); **G02B7/02** (20210101); **H02K41/03** (20060101); **H04N23/68** (20230101); **H10F39/00** (20250101)

U.S. Cl.:

CPC **H04N23/54** (20230101); **H02K41/031** (20130101); **H04N23/687** (20230101); **H10F39/804** (20250101); **G02B7/021** (20130101); **H02K2201/18** (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/176,140, filed Feb. 28, 2023; which is a continuation of U.S. application Ser. No. 17/294,625, filed May 17, 2021, now U.S. Pat. No. 11,622,064, issued Apr. 4, 2023; which is the U.S. national stage application of International Patent Application No. PCT/KR2019/015531, filed Nov. 14, 2019, which claims the benefit under 35 U.S.C. § 119 of Korean Application Nos. 10-2018-0143392, filed Nov. 20, 2018; and 10-2019-0002156, filed Jan. 8, 2019, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] An embodiment relates to a substrate for an image sensor. Specifically, the embodiment relates to a substrate for an image sensor capable of moving relative to a lens barrel and a camera module including the same.

BACKGROUND ART

[0003] In general, camera devices are mounted in mobile communication terminals and portable devices such as MP3 players, as well as electronic devices such as automobiles, endoscopes, and CCTVs. Such camera devices are gradually being developed centering on high pixels, and miniaturization and thinning are in progress. In addition, current camera devices are being changed so that various additional functions can be implemented at low manufacturing cost.

[0004] The camera device as described above includes a lens barrel accommodating a lens, a lens holder coupled to the lens barrel, an image sensor disposed in the lens holder, and a driving substrate on which the image sensor is mounted. At this time, the lens prod provides an image signal of a subject to the image sensor. And the image sensor converts the image signal into an electrical signal.

[0005] Here, the accuracy of the image signal in the camera device is determined according to a focal length defined as the distance between the lens and the image sensor.

[0006] Accordingly, the camera device provides focus compensation or shake compensation by moving the lens barrel relative to the image sensor. That is, the camera device moved the lens barrel accommodating the lens relative to the image sensor in the X-axis, Y-axis, and Z-axis. At this time, the camera device required an elastic member such as at least six springs to move the lens barrel. In addition, the elastic member was bonded to the lens barrel by bonding.

[0007] However, the camera device according to the prior art as described above, by moving the lens barrel, the upper spring plate disposed on the lens barrel, the lower spring plate under the lens barrel, and an elastic wire for fixing the Z-axis (elastic wire) Includes structures such as Accordingly, the module structure of the camera device according to the prior art has a complicated problem.

[0008] In addition, the camera device according to the prior art requires a plurality of elastic members for moving the lens barrel, and there is a problem in that the number of assembling steps

of the plurality of elastic members increases.

DISCLOSURE

Technical Problem

[0009] In the embodiment, it is possible to provide a substrate for an image sensor having a new structure and a camera module including the same.

[0010] In addition, in the embodiment, it is possible to provide an image sensor substrate for an image sensor moving relative to a lens barrel, and a camera module including the same.

[0011] In addition, in the embodiment, a substrate for an image sensor capable of not only movement of the X-axis, Y-axis, and Z-axis, but also tilt correction, and a camera module including the same are provided.

[0012] In addition, in the embodiment, a substrate for an image sensor capable of simplifying a spring structure for providing an auto focus function or a camera shake compensation function, and a camera module including the same can be provided.

[0013] The technical problems to be achieved in the proposed embodiment are not limited to the technical problems mentioned above, and other technical problems not mentioned may be clearly understood by those of ordinary skill in the art to which the proposed embodiment belongs from the following description.

Technical Solution

[0014] A substrate for an image sensor according to an embodiment comprises: a spring plate having an elastic member disposed in a first open region; an insulating layer disposed on the spring plate and including a second open region exposing the first open region; and a conductive pattern portion disposed on the insulating layer, wherein the spring plate comprises a first plate part, and a second plate part disposed to surround the first plate part with the first open region interposed therebetween, and connected to the first plate part by the elastic member, wherein the insulating layer includes a first insulating part disposed on the first plate part, and a second insulating part disposed on the second plate part, wherein the conductive pattern portion, a first lead pattern part disposed on the first insulating part, a second lead pattern part disposed on the second insulating part, and an extension pattern part disposed between the first and second lead pattern parts and floating on the second open region.

[0015] In addition, the substrate comprises a first bonding sheet disposed between the spring plate and the insulating layer and including a third open region exposing the first open region; and a second bonding sheet disposed between the insulating layer and the conductive pattern portion and including a fourth open region exposing the second open region.

[0016] In addition, a length of the extension pattern part is greater than a linear distance between the first lead pattern part and the second lead pattern part.

[0017] In addition, the extension pattern part has a length of 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part.

[0018] In addition, the extension pattern part does not overlap with the elastic member in a vertical direction.

[0019] In addition, the extension pattern part has a tensile strength of 1000 MPa or more.

[0020] In addition, the extension pattern part is formed of a binary alloy of any one of copper (Cu)-nickel (Ni), copper (Cu)-tin (Sn), copper (Cu)-beryllium (Be) and copper (Cu)-cobalt (Co), or a ternary alloy of any one of copper (Cu)-nickel (Ni)-tin (Sn) and copper (Cu)-beryllium (Be)-cobalt (Co).

[0021] In addition, the first lead pattern part, the second lead pattern part, and the extension pattern part include a metal layer disposed on the insulating layer and the second open region, and a plating layer disposed on the metal layer, wherein the plating layer of the extension pattern part is disposed to surround an upper surface, a side surface, and a lower surface of the metal layer of the extension pattern part.

[0022] In addition, a distance between a lower surface of the first lead pattern part and an upper

surface of the spring plate is equal to a distance between a lower surface of the second lead pattern part and an upper surface of the spring plate, and a distance between the lower surface of the extension pattern part and the upper surface of the spring plate is smaller than the distance between the lower surface of the first lead pattern part and the upper surface of the spring plate or the distance between the lower surface of the second lead pattern part and the upper surface of the spring plate.

[0023] In addition, the spring plate includes a plurality of slits formed passing through an upper surface and a lower surface of the first plate part.

[0024] In addition, a plurality of the first lead pattern parts are disposed around an image sensor mounting area on an upper surface of the first insulating part, and wherein each of the second lead pattern part and the extension pattern part includes a plurality of parts to be respectively connected to the plurality of first lead pattern parts.

[0025] In addition, the substrate includes a supporting layer disposed under the lower surface of the second plate part.

[0026] A camera module according to the embodiment comprises a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; an image sensor disposed on the image sensor substrate; a first driving part disposed on a lower surface of the image sensor substrate; and a second driving part disposed in the housing and having an upper surface facing a lower surface of the first driving part, wherein the image sensor substrate comprise: a spring plate having an elastic member disposed in a first open region; an insulating layer disposed on the spring plate and including a second open region exposing the first open region; and a conductive pattern portion disposed on the insulating layer, wherein the spring plate comprises a first plate part, and a second plate part disposed to surround the first plate part with the first open region interposed therebetween, and connected to the first plate part by the elastic member, wherein the insulating layer includes a first insulating part disposed on the first plate part, and a second insulating part disposed on the second plate part, wherein the conductive pattern portion, a first lead pattern part disposed on the first insulating part, a second lead pattern part disposed on the second insulating part, and an extension pattern part disposed between the first and second lead pattern parts and floating on the second open region.

[0027] In addition, the first driving part is disposed under a lower surface of the first plate part, and the second driving part interacts with the first driving part to move the first driving part so that the image sensor substrate is moved relative to the lens barrel in a horizontal direction.

[0028] In addition, the camera module comprises a third driving part having an upper surface facing a side surface of the first driving part on a side surface of the supporting layer, and the third driving part interacts with the first driving part to move the first driving part so that the image sensor substrate is moved relative to the lens barrel in a vertical direction.

[0029] In addition, the extension pattern part has a length of 1.5 to 4 times a linear distance between the first lead pattern part and the second lead pattern part.

[0030] In addition, the extension pattern part does not overlap with the elastic member in a vertical direction.

[0031] On the other hand, the camera device according to the embodiment includes a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; an image sensor disposed on the image sensor substrate; and flexible circuit board including a first connector part disposed in the housing and electrically connected to the image sensor substrate, a second connector part disposed outside the housing and electrically connected to an external device, and a connection part connecting the first and second connectors, wherein the image sensor substrate comprise: a spring plate having an elastic member disposed in a first open region; an insulating layer disposed on the spring plate and including a second open region exposing the first open region; and a conductive pattern portion disposed on the insulating layer, wherein the spring plate comprises a first plate part, and a second plate part

disposed to surround the first plate part with the first open region interposed therebetween, and connected to the first plate part by the elastic member, wherein the insulating layer includes a first insulating part disposed on the first plate part, and a second insulating part disposed on the second plate part, wherein the conductive pattern portion, a first lead pattern part disposed on the first insulating part, a second lead pattern part disposed on the second insulating part, and an extension pattern part disposed between the first and second lead pattern parts and floating on the second open region.

[0032] On the other hand, the substrate for an image sensor according to another embodiment comprises an insulating layer; and a conductive pattern portion disposed on the insulating layer, wherein the insulating layer comprises: a first insulating part; a second insulating part surrounding the first insulating part and spaced apart from the first insulating part with a first open region therebetween, and an extension insulating part disposed in the first open region and connecting the first insulating part and the second insulating part, and the conductive pattern portion comprises: a first lead pattern part disposed on the first insulating part; a second lead pattern part disposed on the second insulating part; and an extension pattern part on the extension insulating part and connecting between the first lead pattern part and the second lead pattern part.

[0033] In addition, the substrate comprises a bonding sheet disposed between the insulating layer and the conductive pattern portion and including a second open region exposing the first open region, wherein the bonding sheet comprises: a first bonding part between the first lead pattern part and the first insulating part; and a second bonding part between the second lead pattern part and the second insulating part; and an extension bonding part between the extension pattern part and the extension insulating part.

[0034] In addition, a length of the extension insulating part is greater than a linear distance of the first open region between the first insulating part and the second insulating part.

[0035] In addition, the extension insulating part has a length of 1.5 to 4 times the linear distance of the first open region between the first insulating part and the second insulating part.

[0036] In addition, the extension pattern part, the extension insulating part, and the extension bonding part overlap each other in a vertical direction.

[0037] In addition, the length of the extension pattern part is greater than a linear distance between the first lead pattern part and the second lead pattern part.

[0038] In addition, the extension pattern part has a length of 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part.

[0039] In addition, the extension insulating part has a line width larger than a line width of the extension pattern part.

[0040] In addition, the extension pattern part has a tensile strength of 1000 MPa or more.

[0041] In addition, the extension pattern part is formed of a binary alloy of any one of copper (Cu)-nickel (Ni), copper (Cu)-tin (Sn), copper (Cu)-beryllium (Be) and copper (Cu)-cobalt (Co), or a ternary alloy of any one of copper (Cu)-nickel (Ni)-tin (Sn) and copper (Cu)-beryllium (Be)-cobalt (Co).

[0042] In addition, the first lead pattern part, the second lead pattern part, and the extension pattern part include a metal layer and a plating layer disposed on upper and side surfaces of the metal layer.

[0043] In addition, a plurality of the first lead pattern parts are disposed around an image sensor mounting area on an upper surface of the first insulating part, and wherein each of the second lead pattern part and the extension pattern part includes a plurality of parts to be respectively connected to the plurality of first lead pattern parts.

[0044] In addition, the substrate comprises a supporting layer disposed under the lower surface of the second insulating part.

[0045] On the other hand, the camera module according to the embodiment comprises a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; an image sensor disposed on the image sensor substrate; a first

driving part disposed on a lower surface of the image sensor substrate; and a second driving part disposed in the housing and having an upper surface facing a lower surface of the first driving part, wherein the image sensor substrate comprise: an insulating layer; and a conductive pattern portion disposed on the insulating layer, wherein the insulating layer comprises: a first insulating part; a second insulating part surrounding the first insulating part and spaced apart from the first insulating part with a first open region therebetween, and an extension insulating part disposed in the first open region and connecting the first insulating part and the second insulating part, and the conductive pattern portion comprises: a first lead pattern part disposed on the first insulating part; a second lead pattern part disposed on the second insulating part; and an extension pattern part on the extension insulating part and connecting between the first lead pattern part and the second lead pattern part.

[0046] In addition, the first driving part is disposed under a lower surface of the first plate part, and the second driving part interacts with the first driving part to move the first driving part so that the image sensor substrate is moved relative to the lens barrel in a horizontal direction.

[0047] In addition, the camera module comprises a third driving part having an upper surface facing a side surface of the first driving part on a side surface of the supporting layer, and the third driving part interacts with the first driving part to move the first driving part so that the image sensor substrate is moved relative to the lens barrel in a vertical direction.

[0048] In addition, the extension pattern part has a length of 1.5 to 4 times a linear distance between the first lead pattern part and the second lead pattern part.

[0049] On the other hand, the camera device according to the embodiment includes a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; an image sensor disposed on the image sensor substrate; and flexible circuit board including a first connector part disposed in the housing and electrically connected to the image sensor substrate, a second connector part disposed outside the housing and electrically connected to an external device, and a connection part connecting the first and second connectors, wherein the image sensor substrate comprise: an insulating layer; and a conductive pattern portion disposed on the insulating layer, wherein the insulating layer comprises: a first insulating part; a second insulating part surrounding the first insulating part and spaced apart from the first insulating part with a first open region therebetween, and an extension insulating part disposed in the first open region and connecting the first insulating part and the second insulating part, and the conductive pattern portion comprises: a first lead pattern part disposed on the first insulating part; a second lead pattern part disposed on the second insulating part; and an extension pattern part on the extension insulating part and connecting between the first lead pattern part and the second lead pattern part, wherein the first connector part includes a pad part disposed on the second insulating part and connected to the second lead pattern part, wherein the extension pattern part has a length of 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part, and wherein the extension insulating part has a line width larger than a line width of the extension pattern part.

Effects of the Invention

[0050] According to an embodiment, in order to implement the OIS and AF functions of the camera module, instead of moving the conventional lens barrel, the image sensor is moved relative to the lens barrel in the X-axis, Y-axis and Z-axis directions. Accordingly, the camera module according to the embodiment may remove a complex spring structure for implementing the OIS and AF functions, and thus the structure may be simplified. In addition, by moving the image sensor according to the embodiment relative to the lens barrel, it is possible to provide a stable structure compared to the conventional structure.

[0051] In addition, according to an embodiment, the extension pattern part electrically connected to the image sensor has a spring structure and is disposed in a floating form on the spring plate.

[0052] In addition, in the insulating layer, an extension insulating part having a spring shape is

disposed in a region vertically overlapping with the extension pattern part. Accordingly, the camera module may elastically support the image sensor more stably and move the image sensor with respect to the lens barrel.

[0053] In addition, the length of the extension pattern part in the embodiment is set to be at least 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part. Accordingly, noise generation can be minimized while improving the mobility of the image sensor substrate.

[0054] In addition, in the embodiment, the width of the extension insulating part is made larger than the width of the extension pattern part, so that the extension pattern part can be stably supported by the extension insulating part, thereby improving operational reliability.

[0055] In addition, according to the embodiment, the elastic member and the extension pattern part are not aligned with each other in a vertical direction, thereby solving an electrical reliability problem that may occur due to contact between the elastic member and the extension pattern part.

Description

DESCRIPTION OF DRAWINGS

[0056] FIG. 1 is a view showing a camera module according to a comparative example.

[0057] FIG. 2 is a view showing a camera device according to a first embodiment.

[0058] FIG. 3 is a view showing a spring plate shown in FIG. 2.

[0059] FIG. 4 is a view showing first and second bonding sheets shown in FIG. 2.

[0060] FIG. 5 is a view showing a first embodiment of the insulating layer shown in FIG. 2.

[0061] FIG. 6 is a view showing a second embodiment of the insulating layer shown in FIG. 2.

[0062] FIG. 7 is a view showing a third embodiment of the insulating layer shown in FIG. 2.

[0063] FIG. 8 is a view showing a pattern part shown in FIG. 2.

[0064] FIG. 9 is a plan view of a substrate for an image sensor according to the first embodiment.

[0065] FIG. 10 is a view showing a connection structure between a flexible circuit board and an image sensor substrate according to the first embodiment.

[0066] FIG. 11 is a view specifically showing a layer structure of a conductive pattern portion according to the first embodiment.

[0067] FIG. 12 is a view showing a modified example of the camera module.

[0068] FIG. 13 is a view showing a X-axis displacement simulation of a spring plate according to the first embodiment.

[0069] FIG. 14 is a view showing a Y-axis displacement simulation of a spring plate according to the first embodiment.

[0070] FIG. 15 is a view showing a Z-axis displacement simulation of a spring plate according to the first embodiment.

[0071] FIG. 16 is a view showing a tilt axis displacement simulation of a spring plate according to the first embodiment.

[0072] FIG. 17 is a view showing a camera device according to a second embodiment.

[0073] FIG. 18 is a view showing an example of the insulating layer shown in FIG. 17.

[0074] FIG. 19 is a view showing a modified example of the insulating layer shown in FIG. 17.

[0075] FIG. 20 is a view showing a second bonding sheet shown in FIG. 17.

[0076] FIG. 21 is a view showing a conductive pattern portion shown in FIG. 17.

[0077] FIG. 22 is a plan view of a substrate for an image sensor according to a second embodiment.

[0078] FIG. 23 is a view showing a connection structure between a flexible circuit board and an image sensor substrate according to a second embodiment.

[0079] FIG. 24 is a view specifically showing a layer structure of a conductive pattern portion

according to a second embodiment.

[0080] FIG. **25** is a view showing a camera device according to a third embodiment.

[0081] FIG. **26A** and FIG. **26B** are views showing a camera device according to a fourth embodiment.

BEST MODE

[0082] Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings.

[0083] However, the spirit and scope of the present invention is not limited to a part of the embodiments described, and may be implemented in various other forms, and within the spirit and scope of the present invention, one or more of the elements of the embodiments may be selectively combined and replaced.

[0084] In addition, unless expressly otherwise defined and described, the terms used in the embodiments of the present invention (including technical and scientific terms may be construed the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs, and the terms such as those defined in commonly used dictionaries may be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art. Further, the terms used in the embodiments of the present invention are for describing the embodiments and are not intended to limit the present invention.

[0085] In this specification, the singular forms may also include the plural forms unless specifically stated in the phrase, and may include at least one of all combinations that may be combined in A, B, and C when described in “at least one (or more) of A (and), B, and C”. Further, in describing the elements of the embodiments of the present invention, the terms such as first, second, A, B, (A, and (b) may be used.

[0086] These terms are only used to distinguish the elements from other elements, and the terms are not limited to the essence, order, or order of the elements. In addition, when an element is described as being “connected”, “coupled”, or “connected” to another element, it may include not only when the element is directly “connected” to, “coupled” to, or “connected” to other elements, but also when the element is “connected”, “coupled”, or “connected” by another element between the element and other elements.

[0087] In addition, when described as being formed or disposed “on (over)” or “under (below)” of each element, the “on (over)” or “under (below)” may include not only when two elements are directly connected to each other, but also when one or more other elements are formed or disposed between two elements. Further, when expressed as “on (over)” or “under (below)”, it may include not only the upper direction but also the lower direction based on one element.

[0088] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0089] FIG. **1** is a view showing a camera module according to a comparative example.

[0090] A camera module having an optical image stabilizer (OIS) function and an Auto Focusing (AF) function requires at least two spring plates.

[0091] The camera module according to the comparative example may have two spring plates. The camera module according to the comparative example requires an elastic member such as at least six springs for the spring plate.

[0092] Referring to FIG. **1**, the camera module according to the comparative example includes an optical system including a lens assembly, an infrared cut-off filter, and a sensor unit. That is, the camera module according to the comparative example includes a lens barrel **10**, a lens assembly **20**, a first elastic member **31**, a second elastic member **32**, a first housing **41**, a housing **42**, an infrared cut-off filter **50**, a sensor unit **60**, a circuit board **80**, and driving parts **71**, **72**, **73**, and **74**.

[0093] In this case, the lens barrel **10** is connected to the first housing **41**. That is, the lens barrel **10** is connected to the first housing **41** via the first elastic member **31**. That is, the lens barrel **10** is connected to the first housing **41** so as to be movable by the first elastic member **31**. In this case,

the first elastic member **31** includes a plurality of springs (not shown). For example, the first elastic member **31** connects between the lens barrel **10** and the first housing **41** at a plurality of points of the lens barrel **10**.

[0094] The second elastic member **32** is connected to the first housing **41** and the second housing **42** accommodating the first housing **41**. The second elastic member **32** fixes the first housing **41** to the second housing **42** so as to be movable. The second elastic member **32** includes a plurality of springs. In detail, the second elastic member **32** includes a plate-shaped spring.

[0095] In this case, the first elastic member **31** moves the lens barrel **10** relative to the sensor unit **60** in a vertical direction (a Z-axis direction) while supporting the lens barrel **10**. To this end, the first elastic member **31** includes at least four springs.

[0096] In addition, the second elastic member **32** moves the lens barrel **10** relative to the sensor unit **60** in a horizontal direction (an X-axis direction and a Y-axis direction) while supporting the lens barrel **10**. To this end, the second elastic member **32** includes at least two springs.

[0097] As described above, in the camera module according to the comparative example, OIS and AF are performed as the lens barrel **10** moves in the X-axis, Y-axis, and Z-axis directions. To this end, the camera module according to the comparative example requires at least six elastic members such as springs. In addition, the camera module according to the comparative example requires two spring plates for supporting the elastic member as described above. Further, the camera module according to the comparative example requires an additional member such as an elastic wire for fixing the Z-axis of the lens barrel **10**. Therefore, the camera module according to the comparative example has a complicated spring structure for moving the lens barrel in the X-axis, Y-axis and Z-axis directions.

[0098] In addition, in the camera module according to the comparative example, it is necessary to manually perform an operation of bonding the respective elastic members in order to couple the elastic member with the lens barrel **10**. Accordingly, the camera module according to the comparative example has a complicated manufacturing process and requires a long manufacturing time.

[0099] In addition, the camera module according to the comparative example provides a tilt function of the lens barrel **10**, but has a structure in which tilt correction of an image is substantially difficult. That is, even though the lens barrel **10** rotates with respect to the sensor unit **60**, an image incident on the sensor unit **60** does not change, and thus the tilt correction of the image is difficult, and further, the tilt function itself is unnecessary.

[0100] Hereinafter, a substrate for an image sensor, a camera module, and a camera device including the same according to an embodiment will be described.

[0101] FIG. **2** is a view showing a camera device according to the first embodiment.

[0102] Referring to FIG. **2**, the camera device according to the first embodiment includes a lens barrel **100**, a lens assembly **200**, a housing **300**, an infrared cut filter part **400**, a driving part **510**, **520**, **530**, and an image sensor substrate **600**, an image sensor **700**, and a flexible circuit board **800**.

[0103] The lens barrel **100** accommodates the lens assembly **200**.

[0104] The lens barrel **100** may include a receiving groove for accommodating the lens assembly **200**. The receiving groove may have a shape corresponding to the lens assembly **200**.

[0105] The lens barrel **100** may have a rectangular tube shape or a cylindrical shape. That is, the outer periphery of the lens barrel **100** may have a rectangular tube shape or a cylindrical shape, but the embodiment is not limited thereto.

[0106] The lens barrel **100** is connected to the housing **300**. The lens barrel **100** is accommodated in the housing **300**. The lens barrel **100** may be coupled to the housing **300** by a separate coupling member (not shown).

[0107] The lens barrel **100** may include an open region in an upper portion thereof. Preferably, the lens barrel **100** may include a light entrance groove that is open to an object side. The light entrance groove may expose the lens assembly **200**. In addition, an image may be incident on the

lens assembly **200** via the light entrance groove.

[0108] The lens assembly **200** is disposed in the lens barrel **100**. The lens assembly **200** is accommodated in the accommodating groove provided in the lens barrel **100**. The lens assembly **200** may be inserted into and fixed to the accommodating groove of the lens barrel **100**. The lens assembly **200** may have a circular outer shape. For example, the lens assembly **200** may have a circular shape when viewed from a top side, but the embodiment is not limited thereto. That is, the lens assembly **200** may have a rectangular shape when viewed from the top side.

[0109] The lens assembly **200** includes a plurality of lenses. For example, the lens assembly **200** may include first to fourth lenses. The first to fourth lenses may be sequentially stacked. In addition, a spacer (not shown) may be interposed between the lenses. The spacer may space a distance between the lenses constant. In the above, the lens assembly **200** has been described as including four lenses, but the embodiment is not limited thereto. For example, the lens assembly **200** may include one to three lenses, or may include five or more lenses.

[0110] The housing **300** accommodates the lens barrel **100**. The housing **300** fixes a position of the lens barrel **100** via a separate fixing member (not shown). That is, according to the comparative example, the lens barrel is coupled to be movable with respect to the housing. Unlike this, in an embodiment, the housing **300** may be firmly fixed via the fixing member such that the lens barrel **100** does not move in the housing **300**. Accordingly, the position of the lens barrel **100** in the housing **300** is always fixed. Accordingly, in the embodiment, since the lens barrel **100** is always fixed at the same position, it is possible to solve a problem of distortion of an optical axis caused by warping of the lens barrel, etc., thereby improving reliability.

[0111] The housing **300** may be formed of plastic or metal. The housing **300** may have a rectangular tube shape.

[0112] The infrared cut-off filter part **400** may be disposed at a lower end of the lens barrel **100**. The infrared cut-off filter part **400** may be fixedly disposed on a separate substrate (not shown), and accordingly, it may be coupled to the lens barrel **100**. The infrared cut-off filter part **400** may block light having an excessive long wavelength flowing into the image sensor **700**.

[0113] The infrared cut-off filter **400** may be formed by alternately depositing titanium oxide and silicon oxide on an optical glass. In this case, thicknesses of the titanium oxide and the silicon oxide constituting the infrared cut-off filter **400** may be appropriately adjusted in order to block infrared rays.

[0114] The driving parts **510**, **520**, and **530** move the image sensor substrate **600** relative to the fixed lens barrel **100**. The driving parts **510**, **520**, and **530** move the image sensor substrate **600** relative to the fixed housing **300**. The driving part **510**, **520**, and **530** move the image sensor substrate **600** relative to the fixed lens assembly **200**.

[0115] To this end, the driving parts **510**, **520**, and **530** may move the image sensor substrate **600** relative to magnetic force. The driving parts **510**, **520**, and **530** may include a first driving part **510**, a second driving part **520**, and a third driving part **530**.

[0116] The first driving part **510** is attached to the image sensor substrate **600**. Preferably, the first driving part **510** may be attached to a lower surface of the image sensor substrate **600**. More preferably, the first driving part **510** may be attached to a lower surface of the insulating layer **610** constituting the image sensor substrate **600**. The first driving part **510** may include a magnet. For example, the first driving part **510** may include a permanent magnet. In this case, the magnet constituting the first driving part **510** may have a plate shape. Accordingly, the first driving part **510** may include an upper surface, a lower surface, and side surfaces.

[0117] The second driving part **520** may be disposed on a bottom surface of the housing **300**. Preferably, the second driving part **520** may be disposed on the bottom surface of the housing **300** overlapped with the image sensor substrate **600** in a vertical direction. The second driving part **520** may include a coil. The second driving part **520** may receive a driving signal and generate a magnetic field according to the driving signal.

[0118] In this case, the first driving part **510** and the second driving part **520** may face each other. That is, the first driving part **510** and the second driving part **520** may be disposed to be overlapped with each other in the vertical direction. The first driving part **510** and the second driving part **520** may be disposed side by side in the horizontal direction. That is, the lower surface of the first driving part **510** and an upper surface of the second driving part **520** may be disposed to face each other. A separation distance between the first driving part **510** and the second driving part **520** may be 50 μm to about 1000 μm , but the embodiment is not limited thereto.

[0119] Magnetic force may be generated between the first driving part **510** and the second driving part **520**. Accordingly, the image sensor substrate **600** may move in the X-axis direction and the Y-axis direction by repulsion force or attraction force generated between the second driving part **520** and the first driving part **510**. In addition, the image sensor substrate **600** may be tilted (or rotated) according to a change in a direction of a current applied to the second driving part **520**. To this end, the second driving part **520** and the first driving part **510** may each include a plurality of magnets and a plurality of coils.

[0120] The third driving part **530** may be attached to the image sensor substrate **600**. Preferably, the third driving part **530** may be attached to a side wall of the image sensor substrate **600**. In detail, the third driving part **530** may be attached to a side surface of the supporting layer **660** of the image sensor substrate **600**. In this case, at least a portion of the third driving part **530** may be disposed to be overlapped with the first driving part **510** in the horizontal direction. The third driving part **530** may be disposed perpendicular to the first driving part **510**. That is, an upper surface of the third driving part **530** may be disposed to face the side surface of the first driving part **510**. Accordingly, the image sensor substrate **600** may move in the Z-axis direction by attraction force and repulsion force between the third driving part **530** and the first driving part **510**.

[0121] The image sensor substrate **600** is a substrate on which the image sensor **700** is mounted. In detail, the image sensor substrate **600** may be driven by the driving parts **510**, **520**, and **530** to move the image sensor **700** in the X, Y, and Z-axis directions. In addition, the image sensor substrate **600** may be driven by the driving parts **510**, **520**, and **530** to tilt the image sensor **700**.

[0122] The image sensor substrate **600** may be disposed to be spaced apart from the bottom surface of the housing **300** by a predetermined distance. In addition, the image sensor substrate **600** may move the mounted image sensor **700** relative to the housing **300**.

[0123] To this end, the image sensor substrate **600** may include a spring plate **610**, a first bonding sheet **620**, an insulating layer **630**, a second bonding sheet **640**, a conductive pattern portion **650**, and a supporting layer **660**.

[0124] The spring plate **610** supports the insulating layer **630** and the conductive pattern portion **650** constituting the image sensor substrate **600**. In addition, the spring plate **610** may move the image sensor **700** disposed on the image sensor substrate **600** in the X-axis, Y-axis and Z-axis directions. To this end, the spring plate **610** may include at least one elastic member. Preferably, the spring plate **610** may include a plurality of elastic members. For example, the spring plate **610** may include four elastic members.

[0125] The spring plate **610** may be formed of a metal material such as stainless steel (STS) or invar, but is not limited thereto. For example, the spring plate **610** may be formed of another metal material such as a spring material having elasticity in addition to the above material. That is, the spring plate **610** may have a certain elastic force. Accordingly, the spring plate **610** may elastically support the image sensor substrate **600** and move the image sensor substrate **600** in the X-axis, Y-axis and Z-axis directions.

[0126] An insulating layer **630** is disposed on the spring plate **610**. In this case, a first bonding sheet **620** may be disposed between the spring plate **610** and the insulating layer **630**. The first bonding sheet **620** may be disposed between the spring plate **610** and the insulating layer **630** to provide adhesion. That is, the first bonding sheet **620** may fix the insulating layer **630** on the spring plate **610**. To this end, the first bonding sheet **620** may be formed of a double-sided adhesive film.

The first bonding sheet **620** may be formed of an epoxy or acrylic adhesive.

[0127] An insulating layer **630** is disposed on the first bonding sheet **620**. That is, the insulating layer **630** may be attached on the spring plate **610** by the first bonding sheet **620**.

[0128] The insulating layer **630** is a substrate for forming the conductive pattern portion **650**. The insulating layer **630** may include all of a printed circuit board, a wiring board, and an insulating substrate made of an insulating material capable of forming the conductive pattern portion **650** on the surface.

[0129] The insulating layers **610** may be rigid or flexible. For example, the insulating layer **610** may include glass or plastic. Specifically, the insulating layer **610** may include a chemically tempered/semi-tempered glass, such as soda lime glass, aluminosilicate glass, etc., a tempered or flexible plastic such as polyimide (PI), polyethylene terephthalate (PET), propylene glycol (PPG), polycarbonate (PC), etc., or sapphire.

[0130] Further, the insulating layer **610** may include an optically isotropic film. For example, the insulating layer **610** may include cyclic olefin copolymer (COC), cyclic olefin polymer (COP), optically isotropic PC, optically isotropic polymethylmethacrylate (PMMA), etc.

[0131] In this case, the insulating layer **610** may be partially bent while having a curved surface. That is, the insulating layer **610** may partially have a plane and may partially be bent while having a curved surface. In addition, the insulating layer **610** may be a flexible substrate having flexibility. Further, the insulating layer **610** may be a curved or bended substrate.

[0132] A conductive pattern portion **650** is disposed on the insulating layer **630**. The conductive pattern portion **650** may include a lead pattern part spaced apart from each other by a predetermined distance on the insulating layer **630**. For example, the conductive pattern portion **650** may include a first lead pattern part connected to the image sensor **700** and a second lead pattern part connected to the flexible circuit board **800**. In addition, the conductive pattern portion **650** may include an extension pattern part connecting the first lead pattern part and the second lead pattern part. The first lead pattern part, the second lead pattern part, and the extension pattern part will be described in detail below.

[0133] The conductive pattern part **650** is a wiring for transmitting an electrical signal, and may be formed of a metal material having high electrical conductivity. For this, the conductive pattern part **640** may be formed of at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn). In addition, the circuit pattern **112** may be formed of paste or solder paste including at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn), which are excellent in bonding strength.

[0134] Preferably, the conductive pattern part **650** serves as a wiring for transmitting an electrical signal, and may be formed of a metallic material having an elastic force movable in the X-axis, Y-axis, and Z-axis directions by interlocking with the elastic member of the spring plate **610**. To this end, the conductive pattern part **650** may be formed of a metal material having a tensile strength of **1000** MPa or more. For example, the conductive pattern part **650** may be a binary alloy or ternary alloy containing copper. For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-nickel (Ni). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-tin (Sn). For example, the conductive pattern part **640** may be a binary alloy of copper (Cu)-beryllium (Be). For example, the conductive pattern part **640** may be a binary alloy of copper (Cu)-cobalt (Co). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-nickel (Ni)-tin (Sn). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-beryllium (Be)-cobalt (Co). In addition, in addition to the metal material, the conductive pattern portion **650** may be formed of an alloy of iron (Fe), nickel (Ni), zinc (Zn), and the like having an elastic force capable of acting as a spring and having good electrical characteristics. Further, the conductive pattern portion **650** may be surface-treated with a plating layer containing a metal material such as gold (Au), silver (Ag), palladium (Pd), and the like,

thereby improving electrical conductivity.

[0135] Meanwhile, the conductive pattern portion **650** may be formed by a general process of manufacturing a printed circuit board, such as an additive process, a subtractive process, a modified semi additive process (MSAP), a semi additive process (SAP), etc., and detailed descriptions thereof will be omitted herein.

[0136] The supporting layer **660** may be a supporting substrate. A third driving part **530** may be mounted on the supporting layer **660**. Accordingly, an electric signal may be applied to the third driving part **530**. The supporting layer **660** may support the spring plate **610**. Preferably, the supporting layer **660** may support the spring plate **610**, the first bonding sheet **620**, the second bonding sheet **640**, the insulating layer **630**, the conductive pattern portion **650** and the image sensor **700** at a position floating from a bottom surface of the housing **300**.

[0137] The image sensor **700** is disposed on the image sensor substrate **600**. Preferably, the image sensor **700** is disposed on the conductive pattern portion **650** of the image sensor substrate **600**. The image sensor **700** may be mounted on the first lead pattern part constituting the conductive pattern portion **650** of the image sensor substrate **600**.

[0138] The image sensor **700** may be composed of a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS) sensor, and may be a device that outputs a photographed object using a photoelectric conversion element and a charge coupled element as an electrical signal. The image sensor **700** may be electrically connected to the conductive pattern portion **650** on the image sensor substrate **600**, and receive an image provided from the lens assembly **200**. In addition, the image sensor **700** may convert the received image into an electrical signal and output the electrical signal. In this case, a signal output through the image sensor **700** may be transmitted to the flexible circuit board **800** via the conductive pattern portion **650**.

[0139] One end of the flexible circuit board **800** may be connected to the image sensor substrate **600**. The flexible circuit board **800** may receive an electrical signal output from the image sensor **700**. The flexible circuit board **800** may have a connector at the other end. A main board (not shown) may be connected to the connector.

[0140] That is, the flexible circuit board **800** may connect the camera module and the main board of the external device. Specifically, the flexible circuit board **800** may connect between the conductive pattern portion **650** of the image sensor substrate **600** of the camera module and the main substrate of the portable terminal.

[0141] To this end, a region of the flexible circuit board **800** is disposed inside the housing **300**, and accordingly, may be connected to the conductive pattern portion **650** of the image sensor substrate **600**.

[0142] Hereinafter, the image sensor substrate **600** according to the first embodiment will be described in more detail.

[0143] FIG. **3** is a view showing the spring plate shown in FIG. **2**.

[0144] Referring to FIG. **3**, the spring plate **610** may elastically support the insulating layer **630** and the conductive pattern portion **650** constituting the image sensor substrate **600**.

[0145] The spring plate **610** may move the image sensor **700** disposed on the image sensor substrate **600** in the X-axis, Y-axis and Z-axis directions.

[0146] To this end, the spring plate **610** may be formed of a material having elasticity. Specifically, the spring plate **610** may include a plurality of elastic members having elastic force. For example, the spring plate **610** may include four elastic members **614**, **615**, **616**, and **617**.

[0147] The spring plate **610** may be formed of a metal material such as stainless steel (STS) or invar, but the embodiment is not limited thereto. For example, the spring plate **610** may be formed of another metal material of a spring material having elastic force in addition to the material. That is, the spring plate **610** may have a predetermined elastic force, and accordingly, the spring plate **610** may move the image sensor substrate **600** in the X-axis, Y-axis, and Z-axis directions while elastically supporting the image sensor substrate **600**.

[0148] The spring plate **610** may include a first plate part **611**, a second plate part **612**, and elastic members **614**, **615**, **616**, and **617**.

[0149] Specifically, the spring plate **610** may have a first plate part **611** disposed at the center thereof. In addition, the second plate part **612** may be disposed surrounding the first plate part **611** at a position spaced apart from the first plate part **611** by a predetermined distance.

[0150] That is, the spring plate **610** may include the first plate part **611** and the second plate part **612**. And, an open region **613** is formed between the first plate part **611** and the second plate part **612**. Preferably, the first plate part **611** and the second plate part **612** may be separated from each other.

[0151] The elastic members **614**, **615**, **616**, and **617** may have one end connected to the first plate part **611** and the other end connected to the second plate part **612**. To this end, the elastic members **614**, **615**, **616**, **617** may include a first elastic member **614**, a second elastic member **615**, a third elastic member **616**, and a fourth elastic member **617**.

[0152] The first elastic member **614** may connect a first corner region of the first plate part **611** and a first corner region of the second plate part **612**. The first elastic member **614** has one end connected to the first corner region of the first plate part **611** and the other end connected to the first corner region of the second plate part **612**. And, the first elastic member **614** may elastically connect them. The first corner region may be a corner portion located at an upper left of each plate.

[0153] The second elastic member **615** may connect a second corner region of the first plate part **611** and a second corner region of the second plate part **612**. The second elastic member **615** has one end connected to the second corner region of the first plate part **611** and the other end connected to the second corner region of the second plate part **612**. And, the second elastic member **615** may elastically connect them. The second corner region may be a corner portion located at an upper right of each plate.

[0154] The third elastic member **616** may connect a third corner region of the first plate part **611** and a third corner region of the second plate part **612**. The third elastic member **616** has one end connected to the third corner region of the first plate part **611** and the other end connected to the third corner region of the second plate part **612**. And, the third elastic member **616** may elastically connect them. The third corner region may be a corner portion located at the lower left of each plate.

[0155] The fourth elastic member **617** may connect a fourth corner region of the first plate part **611** and a fourth corner region of the second plate part **612**. The fourth elastic member **617** has one end connected to the fourth corner region of the first plate part **611** and the other end connected to the fourth corner region of the second plate part **612**. And, the fourth elastic member **617** may elastically connect them. The fourth corner region may be a corner portion located at the lower right of each plate.

[0156] Meanwhile, in the drawings, each of the first plate part **611**, the second plate part **612**, and the elastic members **614**, **615**, **616**, and **617** is illustrated as being configured separately, but is not limited thereto. That is, the first plate part **611**, the second plate part **612**, and the elastic members **614**, **615**, **616**, and **617** may be integrally formed with each other. That is, the spring plate **610** may form an open region **613** on the plate of the plate-like member in the remaining portions except for portions corresponding to the elastic members **614**, **615**, **616**, and **617**.

[0157] Such the spring plate **610** may have a thickness of 10 μm to 100 μm . For example, the spring plate **610** may have a thickness of 20 μm to 70 μm . For example, the spring plate **610** may have a thickness of 40 μm to 50 μm . When the thickness of the spring plate **610** exceeds 100 μm , a thickness of the image sensor substrate **600** may increase. In addition, when the thickness of the spring plate **610** is smaller than 10 μm , a stress generated during movement of the image sensor substrate **600** may not be sufficiently maintained. Preferably, the spring plate **610** is set to have a thickness of 50 μm +10 μm to maintain a stress of 600 MPa or more.

[0158] In addition, the elastic members **614**, **615**, **616**, and **617** have a length equal to or greater

than a predetermined level. When the length of the elastic members **614**, **615**, **616**, and **617** is too long, there is a problem that a volume of the spring plate **610** becomes large, and when the length of the elastic members **614**, **615**, **616**, and **617** is too short, it may not be possible to stably and elastically support the image sensor substrate **600**. Therefore, the elastic members **614**, **615**, **616**, and **617** have a length of 50 μm to 100 μm . At this time, the length of the elastic members **614**, **615**, **616**, and **617** is larger than the width of the open region **613**. That is, the elastic members **614**, **615**, **616**, and **617** may be formed to have a shape of a plurality of bent springs in the open region **613**. [0159] In addition, a plurality of slits **618** are formed in the first plate part **611**. The plurality of slits **618** may be spaced apart from each other by a predetermined distance on the first plate part **61**. The plurality of slits **618** may be formed for reducing a weight of the spring plate **610**. In addition, the plurality of slits **618** may be formed for flatness of the spring plate **610**. That is, the insulating layer **630**, the conductive pattern portion **650**, and the image sensor **700** are disposed on the first plate part **611**. The first driving part is disposed below the first plate part **611**. In this case, flatness of the first driving part or the image sensor **700** directly affects the reliability of the camera module, and the image quality may be deteriorated as the flatness is worsen. Therefore, in the embodiment, not only the weight of the spring plate **610** may be reduced but also the flatness may be maintained by forming the plurality of slits **618** in the first plate part **611**, thereby improving overall reliability of the camera module.

[0160] FIG. 4 is a view showing first and second bonding sheets shown in FIG. 2.

[0161] Referring to FIG. 4, a first bonding sheet **620** is disposed on the spring plate **610**. In this case, the first bonding sheet **620** has a planar shape corresponding to the planar shape of the spring plate **610**.

[0162] That is, the first bonding sheet **620** includes a first bonding part **621** disposed on the first plate part **611** of the spring plate **610** and a second bonding part **622** disposed on the second plate part **612** of the spring plate **610**. In addition, the first bonding sheet **620** may include an open region **623** between the first bonding part **621** and the second bonding part **622**.

[0163] The second bonding part **622** is disposed surrounding the first bonding part **621** at a position spaced apart from the first bonding part **621** by a predetermined distance. In this case, the second bonding part **622** does not directly contact the first bonding part **621**. Accordingly, the first bonding part **621** and the second bonding part **622** may be separated from each other through the open region **623**.

[0164] Meanwhile, the open region **623** of the first bonding sheet **620** may overlap the open region **613** of the spring plate **610** in a vertical direction. Preferably, a planar area of the open region **623** of the first bonding sheet **620** may be the same as a planar area of the open region **613** of the spring plate **610**. In addition, the first bonding sheet **620** does not overlap with the elastic members **614**, **615**, **616**, **617** of the spring plate **610** in the vertical direction. Accordingly, the first plate part **611** and the second plate part **612** may be covered by the first bonding sheet **620**, and the elastic members **614**, **615**, **616**, and **617** may be exposed by the open area **623** of the first bonding sheet **620**.

[0165] The first bonding sheet **620** may be formed of a double-sided adhesive film. The first bonding sheet **620** may be formed of an epoxy or acrylic adhesive or a thermosetting adhesive film.

[0166] The first bonding sheet **620** may have a thickness of 25 μm .

[0167] FIG. 5 is a view showing a first embodiment of the insulating layer shown in FIG. 2, FIG. 6 is a view showing a second embodiment of the insulating layer shown in FIG. 2, and FIG. 7 is a view showing a third embodiment of the insulating layer shown in FIG. 2.

[0168] Referring to FIG. 5, the insulating layer **630** is disposed on the first bonding sheet **620**. The insulating layer **630** may have a planar shape corresponding to the planar shape of the first bonding sheet **620**.

[0169] That is, the insulating layer **630** may include a first insulating part **631** disposed on the first bonding part **621** and a second insulating part **632** disposed on the second bonding part **622**. In

addition, the insulating layer **630** may include an open region **633** between the first insulating part **631** and the second insulating part **632**.

[0170] The second insulating part **632** is disposed surrounding the first insulating part **631** at a position spaced apart from the first insulating part **631** by a predetermined distance. In this case, the second insulating part **632** does not directly contact the first insulating part **631**. Accordingly, the first insulating part **631** and the second insulating part **632** may be separated from each other through the open region **633**.

[0171] Meanwhile, the open region **633** of the insulating layer **630** may overlap the open region **613** of the spring plate **610** in a vertical direction. Preferably, a planar area of the open region **633** of the insulating layer **630** may be the same as the planar area of the open region **613** of the spring plate **610**. In addition, the insulating layer **630** does not overlap with the elastic members **614**, **615**, **616**, **617** of the spring plate **610** in the vertical direction. That is, the elastic members **614**, **615**, **616**, and **617** may be exposed through the open region **623** of the first bonding sheet **620** and the open region **633** of the insulating layer **630**.

[0172] The insulating layer **630** may have a thickness of 20 μm to 100 μm . For example, the insulating layer **630** may have a thickness of 25 μm to 50 μm . For example, the insulating layer **630** may have a thickness of 30 μm to 40 μm . When the thickness of the insulating layer **630** exceeds 100 μm , the overall thickness of the image sensor substrate **600** may increase. When the thickness of the insulating layer **630** is less than 20 μm , it may be difficult to dispose the image sensor **700**. When the thickness of the insulating layer **630** is less than 20 μm , it may be weak against heat/pressure, or the like in a process of mounting the image sensor, and thus it may be difficult to mount the image sensor **700** stably.

[0173] Meanwhile, a second bonding sheet **640** is disposed on the insulating layer **630**.

[0174] The second bonding sheet **640** has the same structure as the first bonding sheet **620**. Compared to the first bonding sheet **620**, the second bonding sheet **640** differs only in an arrangement position. Accordingly, a detailed description of the second bonding sheet **640** will be omitted. Meanwhile, although not shown in the drawing, the second bonding sheet **640** also includes an open area like the first bonding sheet **620** or the insulating layer **630**, and accordingly, the elastic members **614**, **615**, **616**, **617** is exposed.

[0175] Meanwhile, in the insulating layer **630** in the first embodiment, the first insulating part and the second insulating part are separated from each other. That is, the open region **633** of the insulating layer **630** is disposed to surround the entire circumference of the first insulating portion **631**.

[0176] On the other hand, the open region **633** of the insulating layer **630** may be disposed partially surrounding the first insulating part **631**.

[0177] FIG. 6, the insulating layer **630A** in the second embodiment may include an insulating part **624** and a plurality of open holes **625A**, **625B**, **625C**, and **625D** disposed in the insulating part **624**.

[0178] That is, in the first embodiment, the open region has a shape of one closed loop that is entirely disposed around the first insulating part. On the other hand, in the second embodiment, the plurality of open holes **625A**, **625B**, **625C**, and **625D** are disposed at positions spaced apart from each other by a predetermined interval. Accordingly, the insulating part **624** of the insulating layer **630A** is not separated. In this case, the plurality of open holes **625A**, **625B**, **625C**, and **625D** may be formed on a region overlapping the open region **613** of the spring plate **610**. At this time, a connection portion between the plurality of open holes **625A**, **625B**, **625C**, **625D** is positioned on a region overlapping in the vertical direction with the elastic members **614**, **615**, **616**, **617** of the spring plate **610**. According to this embodiment, the elastic support force of the image sensor substrate **600** may be improved by increasing the strength of the insulating layer **630A**.

[0179] Also, referring to FIG. 7, the insulating layer **630B** according to the third embodiment may include a first insulating part **626**, a second insulating part **627**, and an elastic insulating part **628**.

[0180] The elastic insulating part **628** may be integrally formed with the first insulating part **626**

and the second insulating part **627**. That is, when an open region is formed in the insulating layer **630B**, the entire periphery of the first insulating part **626** is not opened, that is, a spring-like pattern may remain around the first insulating part **626**. Accordingly, the elastic insulating part **628** of the embodiment may be formed. The elastic insulating part **628** may be disposed on a region vertically overlapping the elastic members **614**, **615**, **616**, **617** of the spring plate **610**. According to such an embodiment, the elasticity of the image sensor substrate **600** may be improved by increasing the strength of the insulating layer **630B**.

[0181] FIG. **8** is a view showing a pattern part shown in FIG. **2**.

[0182] Referring to FIG. **8**, the conductive pattern portion **650** may be disposed on the insulating layer **630** with a specific pattern. The conductive pattern portion **650** includes a first conductive pattern part **651** disposed on a first region of the insulating layer **630**, a second conductive pattern part **652** disposed on a second region of the insulating layer **630**, a third conductive pattern part **653** disposed on a third region of the insulating layer **630**, and a fourth conductive pattern part **654** disposed on a fourth region of the insulating layer **630**.

[0183] The first conductive pattern part **651** may be disposed on a left side of an upper surface of the insulating layer **630**. That is, the first conductive pattern part **651** may be disposed on a left region of the first insulating part **631**, a left region of the second insulating part **632**, and a left region of the open region **633**.

[0184] The second conductive pattern part **652** may be disposed in a right region of the upper surface of the insulating layer **630**. That is, the second conductive pattern part **652** may be disposed on a right region of the first insulating part **631**, a right region of the second insulating part **632**, and a right region of the open region **633**.

[0185] The third conductive pattern part **653** may be disposed in an upper region of the upper surface of the insulating layer **630**. That is, the third conductive pattern part **653** may be disposed on an upper region of the first insulating part **631**, an upper region of the second insulating part **632**, and an upper region of the open region **633**.

[0186] The fourth conductive pattern part **654** may be disposed in a lower region of the upper surface of the insulating layer **630**. That is, the fourth conductive pattern part **654** may be disposed on a lower region of the first insulating part **631**, a lower region of the second insulating part **632**, and a lower region of the open region **633**.

[0187] As described above, the conductive pattern portion **650** are disposed on different regions, and accordingly, the elastic support force in the movement of the image sensor substrate **600** may be increased. That is, when the conductive pattern portion **650** is intensively disposed only in a specific area, reliability in moving the image sensor substrate **600** in a specific direction may be reduced. For example, when the conductive pattern portion **650** includes only the first and second conductive pattern parts, there is no problem with the movement of the image sensor substrate **600** in the X-axis direction, stability may be deteriorated when the image sensor substrate **600** moves in the Y-axis direction. In addition, in this case, the conductive pattern portion **650** may be disconnected according to the periodic movement of the image sensor substrate **600**. Accordingly, in the embodiment, the conductive pattern portion **650** is distributed in each of the four regions as described above, so that the image sensor substrate **600** can be stably moved in the X-axis, Y-axis, and Z-axis.

[0188] Meanwhile, the conductive pattern portion **650** may include a first lead pattern part **655** connected to the image sensor **700** and a second lead pattern part **656** connected to the flexible circuit board **800**. In addition, the conductive pattern portion **650** may include an extension pattern part **657** connecting the first lead pattern part **655** and the second lead pattern part **656** to each other.

[0189] The first lead pattern part **655** is disposed on the first insulating part **631** of the insulating layer **630**. The first lead pattern part **655** may be disposed in an outer region of the first insulating part **631**. That is, the image sensor mounting area on which the image sensor **700** is mounted may

be formed in the first insulating part **631**. In this case, the image sensor mounting area may be a central area of the first insulating part **631**. Accordingly, the first lead pattern part **655** may be disposed around the image sensor mounting area of the first insulating part **631**.

[0190] The second lead pattern part **656** is disposed on the second insulating part **632** of the insulating layer **630**. The second lead pattern parts **656** may be respectively disposed on the second insulating part **632**. In this case, the first lead pattern part **655** and the second lead pattern part **656** may be disposed to face each other on the first insulating part **631** and the second insulating part **632**. That is, the number of first lead pattern parts **655** may be the same as the number of second lead pattern parts **656**. In addition, the first lead pattern parts **655** may be disposed to face the second lead pattern part **656**, respectively.

[0191] Meanwhile, the extension pattern part **657** may be disposed between the first lead pattern part **655** and the second lead pattern part **656**.

[0192] The extension pattern part **657** may have one end connected to the first lead pattern part **655** and the other end connected to the second lead pattern part **656** facing the first lead pattern part **655**.

[0193] In this case, the extension pattern part **657** may be disposed on the open region of the insulating layer **630**. Accordingly, the extension pattern part **657** may be disposed to float in the open region of the insulating layer, the open region of the first bonding sheet, the open region of the second bonding sheet, and the open region of the spring plate **610**. Here, the meaning that the extension pattern part **657** is floating means that there is no support part supporting the extension pattern part **657** at the bottom. Accordingly, the extension pattern part **657** may be disposed in a state floating in the air.

[0194] A length of the extension pattern part **657** is greater than a linear distance between the first lead pattern part **655** and the second lead pattern part **656**. That is, the extension pattern part **657** may be formed to have a structure that is bent a plurality of times between the first lead pattern part **655** and the second lead pattern part **656**. Preferably, the extension pattern part **657** may be formed in a spring shape between the first lead pattern part **655** and the second lead pattern part **656**.

[0195] At this time, the extension pattern part **657** may be formed by etching to have the above shape through an additive process, a subtractive process, a modified semi-additive process (MSAP), a semi-additive process (SAP), and the like. Preferably, the extension pattern part **657** may be formed simultaneously with the first lead pattern part **655** and the second lead pattern part **656**. More preferably, the extension pattern part **657** may be integrally formed with the first lead pattern part **655** and the second lead pattern part **656**.

[0196] Meanwhile, the thickness of the conductive pattern portion **650** including the extension pattern part **657**, the first lead pattern part **655**, and the second lead pattern part **656** may be 10 μm to 50 μm . For example, the thickness of the conductive pattern portion **650** may be 30 μm to 40 μm . In this case, when the thickness of the conductive pattern portion **650** is less than 10 μm , the conductive pattern portion **650** may be broken when the image sensor substrate **600** moves. In addition, when the thickness of the conductive pattern portion **650** is greater than 50 μm , elastic force of the extension pattern part **657** may be lowered, thereby hindering the mobility of the image sensor substrate **600**. Accordingly, in the embodiment, the thickness of the conductive pattern portion **650** is set to be 35 $\mu\text{m} \pm 5 \mu\text{m}$ such that the image sensor substrate **600** may be stably moved.

[0197] In addition, the length of the extension pattern part **657** is set to have at least 1.5 times the linear distance between the first lead pattern part **655** and the second lead pattern part **656**. In addition, the length of the extension pattern part **657** is set to be 20 times or less of the linear distance between the first lead pattern part **655** and the second lead pattern part **656**. Preferably, the length of the extension pattern part **657** is set to be less than 4 times the linear distance between the first lead pattern part **655** and the second lead pattern part **656**.

[0198] The linear distance between the first lead pattern part **655** and the second lead pattern part

656 may be 1.5 mm.

[0199] In this case, when the length of the extension pattern part **657** is smaller than 1.5 times the linear distance between the first lead pattern part **655** and the second lead pattern part, the mobility of the image sensor substrate **600** may be reduced due to the decrease in the elastic force of the extension pattern part **657**. In addition, when the length of the extension pattern part **657** is greater than 20 times the linear distance, a resistance increases as a signal transmission distance is increased by the extension pattern part **657**, and accordingly, noise may be included in a signal transmitted via the extension pattern part **657**. Accordingly, in order to minimize noise generation, the length of the extension pattern part **657** is set to be 4 times or less the linear distance between the first lead pattern part **655** and the second lead pattern part **656**.

[0200] In addition, the length of the extension pattern part **657** is smaller than the lengths of the elastic members **614**, **615**, **616**, and **617**.

[0201] Meanwhile, the conductive pattern portion **650** as described above is a wiring for transmitting an electrical signal, and may be formed of a metal material having high electrical conductivity. For this, the conductive pattern portion **650** may be formed of at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn). In addition, the conductive pattern portion **650** may be formed of paste or solder paste including at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn), which are excellent in bonding strength.

[0202] Preferably, the conductive pattern portion **650** serves as a wiring for transmitting an electrical signal, and may be formed of a metallic material having an elastic force movable in the X-axis, Y-axis, and Z-axis directions by interlocking with the elastic member. To this end, the conductive pattern portion **650** may be formed of a metal material having a tensile strength of 1000 MPa or more. For example, the conductive pattern portion **650** may be a binary alloy or ternary alloy containing copper. For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-nickel (Ni). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-tin (Sn). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-beryllium (Be). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-cobalt (Co). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-nickel (Ni)-tin (Sn). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-beryllium (Be)-cobalt (Co). In addition, in addition to the metal material, the conductive pattern portion **650** may be formed of an alloy of iron (Fe), nickel (Ni), zinc (Zn), and the like having an elastic force capable of acting as a spring and having good electrical characteristics. Further, the conductive pattern portion **650** may be surface-treated with a plating layer containing a metal material such as gold (Au), silver (Ag), palladium (Pd), and the like, thereby improving electrical conductivity.

[0203] FIG. **9** is a plan view of a substrate for an image sensor according to the first embodiment.

[0204] Referring to FIG. **9**, the first bonding sheet **620** is disposed on the spring plate **610**. In addition, the insulating layer **630** is disposed on the first bonding sheet **620**. In addition, the second bonding sheet **630** is disposed on the insulating layer **630**. In addition, the conductive pattern portion **650** is disposed on the second bonding sheet **630**. In this case, the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630** each include an open region, and each of the open regions may overlap each other in a vertical direction. Therefore, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** disposed at a lowermost portion may be exposed through the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet.

[0205] The extension pattern part **657** of the conductive pattern portion **650** is also disposed on the open region of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630**.

[0206] That is, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** and the extension pattern part **657** of the conductive pattern portion **650** are disposed in the open regions of the spring plate **610**, the first bonding sheet **620**, and the insulating layer **630**, and the second bonding sheet **630**.

[0207] In this case, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** and the extension pattern part **657** of the conductive pattern portion **650** do not overlap each other in the vertical direction. For example, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** are disposed in a first region of the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, the second bonding sheet **630**. And, the extension pattern part **657** of the conductive pattern portion **650** is disposed in a second region of the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630**. The first region of the open regions may be a corner region of the open regions, and the second region of the open regions may be a remaining area except for the corner region.

[0208] That is, the image sensor substrate **600** not only moves in the X-axis direction and the Y-axis direction, but also moves in the Z-axis direction. At this time, there is a difference between the elastic modulus of the elastic members **614**, **615**, **616**, and **617** and the elastic modulus of the extension pattern part **657**. Accordingly, when the image sensor substrate **600** moves in the Z-axis direction, the movement distances of the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** appear different from each other. Accordingly, when the image sensor substrate **600** moves in the Z-axis direction, the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** may contact each other. Accordingly, a problem (for example, a short) may occur in electrical reliability. Accordingly, in the embodiment, as described above, the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** are not aligned with each other in the vertical direction. Thereby, the electrical reliability problem can be solved.

[0209] FIG. **10** is a view showing a connection structure between a flexible circuit board and an image sensor board according to the first embodiment.

[0210] Referring to FIG. **10**, the flexible circuit board **800** electrically connects the image sensor substrate **600** and an external main substrate (not shown) to each other.

[0211] One end of the flexible circuit board **800** may be connected to the image sensor substrate **600**. The flexible circuit board **800** may receive an electrical signal output from the image sensor **700**. The flexible circuit board **800** may have a connector **810** at the other end. A main board (not shown) may be connected to the connector **810**.

[0212] That is, the flexible circuit board **800** may connect the camera module and the main board of the external device. Specifically, the flexible circuit board **800** may connect between the conductive pattern portion **650** of the image sensor substrate **600** of the camera module and the main board of a portable terminal.

[0213] To this end, an area of the flexible circuit board **800** is disposed inside the housing **300**, and accordingly, may be connected to the conductive pattern portion **650** of the image sensor substrate **600**.

[0214] That is, the flexible circuit board **800** may include a first connector part **801**, a second connector part **803**, and a connection part **802**.

[0215] The first connector part **801** may be disposed inside the housing **300**. The first connector part **801** may include a plurality of pads **804**, **805**, **806**, and **807** connected to the conductive pattern portion **650**.

[0216] The first connector part **801** may be electrically connected to the second lead pattern part **656** of the conductive pattern portion **650**. That is, the plurality of pads **804**, **805**, **806**, and **807** of the first connector part **801** may be electrically connected to the second lead pattern part **656**.

[0217] To this end, the first connector part **801** may include a first pad part **804** connected to the second lead pattern part **656** of the first conductive pattern part **651**. In addition, the first connector part **801** may include a second pad part **805** connected to the second lead pattern part **656** of the

second conductive pattern part **652**. In addition, the first connector part **801** may include a third pad part **806** connected to the second lead pattern part **656** of the third conductive pattern part **653**. In addition, the first connector part **801** may include a fourth pad part **807** connected to the second lead pattern part **656** of the fourth conductive pattern part **654**.

[0218] In this case, the first connector part **801** has a shape corresponding to the second insulating part **632** of the insulating layer **630**. Accordingly, the first connector part **801** may be disposed surrounding an upper area of the second lead pattern part **656** of the conductive pattern portion **650**, and the plurality of pads **804**, **805**, **806**, and **807** may be disposed on a lower surface of the first connector part **801**.

[0219] The connection part **802** connects the first connector part **801** and the second connector part **803** to each other. A part of the connection part **802** may be disposed in the housing **300**, and may be extended therefrom to be exposed to the outside of the housing **300**.

[0220] The second connector part **803** may include a connector **810** connected to the main board of the terminal.

[0221] Meanwhile, the image sensor **1700** may be attached on the first insulating part **1631** of the insulating layer **1610**.

[0222] In this case, an electrode **710** of the image sensor **700** faces upward and may be attached to the first insulating part **631**. In addition, a connection member **720** such as a metal wire is formed between the electrode **710** of the image sensor **700** and the first lead pattern part **655**. The connection member **720** may electrically connect the electrode of the image sensor and the first lead pattern part.

[0223] FIG. **11** is a diagram specifically illustrating a layer structure of a conductive pattern portion according to the first embodiment.

[0224] Referring to FIG. **11**, the conductive pattern portion **650** is disposed on the second bonding sheet **640** disposed on the insulating layer **630**. In this case, the conductive pattern portion **650** includes the first lead pattern part **655** disposed on the first insulating part **631** of the insulating layer **630**, the second lead pattern part **656** on the second insulating part **632**, and the extension pattern part **657** connecting therebetween.

[0225] In this case, each of the first lead pattern part **655**, the second lead pattern part **656**, and the extension pattern part **657** may include a metal layer **650A** and a plating layer **650B**.

[0226] The metal layer **650A** may be disposed on the second bonding sheet **640**. That is, the metal layer **650A** is disposed on the second bonding sheet **640** to configure the first lead pattern part **655**, the second lead pattern part **656**, and the extension pattern part **657**, respectively.

[0227] The plating layer **650B** may be disposed on the metal layer **650A**. Preferably, the plating layer **650B** may be a surface treatment layer disposed on the metal layer **650A**.

[0228] The plating layer **650B** includes any one of Ni/Au alloy, gold (Au), electroless nickel immersion gold (ENIG), Ni/Pd alloy, and organic compound plating (Organic Solderability Preservative, OSP).

[0229] In this case, the plating layers **650B** constituting the first lead pattern part **655** and the second lead pattern part **656** may correspond to each other. Alternatively, the plating layer **650B** constituting the extension pattern part **657** may have a thickness different from that of the plated layer **650B** constituting the first lead pattern part **655** and the second lead pattern part **656**.

[0230] That is, the plating layer **650B** of the first lead pattern part **655** and the second lead pattern part **656** may be selectively formed only on upper and side surfaces of the corresponding metal layer **650A**. In contrast, the metal layer **650A** constituting the extension pattern part **657** is disposed in a floating state on the above-described open region. Accordingly, the plating layer **650B** of the extension pattern part **657** may be formed to surround the entire surface of the corresponding metal layer **650A**. That is, the plating layer **650B** of the extension pattern part **657** may be formed to surround the upper, side, and lower surfaces of the metal layer **650A** of the extension pattern part **657**.

[0231] Accordingly, the lower surface of the first lead pattern part **655** and the upper surface of the spring plate **610** may be spaced apart by a first distance. In addition, the lower surface of the second lead pattern part **656** and the upper surface of the spring plate **610** may be spaced apart by the first distance. In addition, the lower surface of the extension pattern part **657** and the spring plate **610** may be spaced apart by a second distance smaller than the first distance due to the plating layer.

[0232] Meanwhile, the thickness of the plating layer **650B** may be 0.3 μm to 1 μm . For example, the thickness of the plating layer **650B** may be 0.3 μm to 0.7 μm . The thickness of the plating layer **650B** may be 0.3 μm to 0.5 μm .

[0233] FIG. **12** is a diagram showing a modified example of the camera module.

[0234] Referring to FIG. **12**, the image sensor may be attached to the image sensor substrate **600** in a flip chip bonding method.

[0235] That is, after the image sensor in the previous embodiment is attached on the insulating layer **630**, the first lead pattern part **655** and the electrode **710** of the image sensor **700** are mutually connected by wire bonding through a connection member.

[0236] Alternatively, the image sensor may be mounted on the package substrate, and accordingly, the image sensor may be mounted on the image sensor substrate **600** on the insulating layer **630** in a flip chip bonding method.

[0237] The package substrate **900** may include an insulating layer **910**, a circuit pattern **920**, a via **930**, an image sensor **940**, a connection member **950**, a protection member **960**, and an adhesive member **970**.

[0238] The image sensor **940** may be electrically connected to the circuit patterns **920** respectively disposed on the upper and lower surfaces of the insulating layer **910** through the connection member **950**. Further, the via **930** may electrically connect the circuit patterns **920** disposed on the upper and lower surfaces of the insulating layer **910** to each other.

[0239] The protection member **960** may be disposed on the lower surface of the insulating layer **910** to protect the lower surface of the insulating layer **910** and expose at least a part of the circuit pattern **920**. In addition, an adhesive member **970** may be disposed on the circuit pattern exposed through the protection member **960**. The adhesive member **970** may be a solder ball. The adhesive member **970** may contain materials of heterogeneous components in solder. The solder may be composed of at least one of SnCu, SnPb, and SnAgCu. In addition, the material of the heterogeneous component may include any one of Al, Sb, Bi, Cu, Ni, In, Pb, Ag, Sn, Zn, Ga, Cd, and Fe.

[0240] Meanwhile, in a state in which the package substrate **900** as described above is manufactured, the adhesive member **970** may be connected to the first lead pattern part **655**.

[0241] FIG. **13** is a view showing a X-axis displacement simulation of a spring plate according to the first embodiment, FIG. **14** is a view showing a Y-axis displacement simulation of a spring plate according to the first embodiment, FIG. **15** is a view showing a Z-axis displacement simulation of a spring plate according to the first embodiment, and FIG. **16** is a view showing a tilt axis displacement simulation of a spring plate according to the first embodiment.

[0242] Referring to FIGS. **13** and **16**, reliability in the X-axis, Y-axis, Z-axis and tilt operation of the image sensor substrate **600** was evaluated by the spring plate **610** according to the embodiment. As a result of this, it was confirmed that the image sensor substrate **600** was stably supported for all movement directions, and it was confirmed that sufficient stress was maintained.

[0243] The results of the X-axis displacement simulation according to the embodiment are shown in Table 1.

TABLE-US-00001 TABLE 1 condition data unit Force 10 mN displacement 3.23 mm stress 409 MPa spring constant 3.10 mM/mm

[0244] The results of the Y-axis displacement simulation according to the embodiment are shown in Table 2.

TABLE-US-00002 TABLE 2 condition data unit Force 10 mN displacement 3.81 mm stress 439 MPa spring constant 2.62 mM/mm

[0245] The results of the Z-axis displacement simulation according to an embodiment are shown in Table 3.

TABLE-US-00003 TABLE 3 condition data unit Force 10 mN displacement 12.12 mm stress 663 MPa spring constant 0.83 mM/mm

[0246] The tilt displacement simulation results according to the embodiment are shown in Table 4 below.

TABLE-US-00004 TABLE 4 condition data unit Force 5×2 (long axis) mN displacement 22.4(1.26) °(mm) stress 237 MPa

[0247] FIG. 17 is a view showing a camera device according to an embodiment.

[0248] Referring to FIG. 17, the camera device according to the embodiment includes a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter part **1400**, a driving part **1510**, **1520**, **1530**, an image sensor substrate **1600**, an image sensor **1700**, and a flexible circuit board **1800**.

[0249] In the following description, a detailed description of the configuration substantially the same as that of FIG. 2 will be omitted.

[0250] Here, the lens barrel **110**, the lens assembly **1200**, the housing **1300**, the infrared cut filter part **1400**, the driving part (**1510**, **1520**, **1530**), the image sensor **1700**, and the flexible circuit board **1800** are substantially the same as the corresponding configuration in FIG. 2, and a detailed description thereof will be omitted.

[0251] The image sensor substrate **1600** is a substrate on which the image sensor **1700** is mounted. Specifically, the image sensor substrate **1600** may be driven by the driving parts **1510**, **1520**, and **1530** to move the image sensor **1700** in the X-axis, Y-axis and Z-axis directions. In addition, the image sensor substrate **1600** may be driven by the driving parts **1510**, **1520**, and **1530** to tilt the image sensor **1700**.

[0252] The image sensor substrate **1600** may be disposed to be spaced apart from a bottom surface of the housing **1300** at a predetermined interval. In addition, the image sensor substrate **1600** may move the image sensor **1700** relative to the housing **1300**.

[0253] To this end, the image sensor substrate **1600** may include an insulating layer **1610**, a first bonding sheet **1620**, a conductive pattern portion **1640**, and a supporting layer **1650**. In addition, according to an embodiment, the image sensor substrate **1600** may further include a second bonding sheet **1630**. That is, the second bonding sheet **1630** may be selectively disposed between the insulating layer **1610** and the conductive pattern portion **1640**. If the second bonding sheet is included, the plane area of the second bonding sheet in one embodiment may be the same as the plane area of the insulating layer **1610**, and the plane area of the second bonding sheet in another embodiment may be the same as the plane area of the conductive pattern portion **1640**. This will be described with reference to FIGS. 25 and 26A and 26B.

[0254] The insulating layer **1610** may support the conductive pattern portion **1640** and move the image sensor **1700** disposed on the image sensor substrate **1600** in the X-axis, Y-axis, and Z-axis directions. To this end, the insulating layer **1610** may include an elastic region having a certain elastic force. Preferably, the insulating layer **1610** may include a plurality of elastic regions. For example, the insulating layer **1610** may include four elastic regions.

[0255] That is, at least one region of the insulating layer **1610** may have a certain elastic force, Accordingly, the image sensor substrate **1600** may be moved in the X-axis, Y-axis, and Z-axis directions while elastically supporting the image sensor substrate **1600**.

[0256] A conductive pattern portion **1640** is disposed on the insulating layer **1610**. The conductive pattern portion **1640** may include a lead pattern part spaced apart from each other by a predetermined interval on the insulating layer **1610**. For example, the conductive pattern portion **1640** may include a first lead pattern part connected to the image sensor **1700** and a second lead

pattern part connected to the flexible circuit board **1800**. In addition, the conductive pattern portion **1650** may include an extension pattern part connecting the first lead pattern part and the second lead pattern part. The first lead pattern part, the second lead pattern part, and the extension pattern part will be described in detail below.

[0257] In addition, the insulating layer **1610** may include a first insulating part in which the first lead pattern part is disposed, a second insulating part in which the second lead pattern part is disposed, and an extension insulating part in which the extension pattern part is disposed. The first insulating part, the second insulating part, and the extension insulating part will be described in more detail below.

[0258] Meanwhile, a second bonding sheet **1630** may be disposed between the insulating layer **1610** and the conductive pattern portion **1640**. The second bonding sheet **1630** may provide adhesive force between the insulating layer **1610** and the conductive pattern portion **1640**. In this case, the second bonding sheet **1630** may have a shape corresponding to the insulating layer **1610**. To this end, the second bonding sheet **1630** may include an elastic region having an elastic force corresponding to the insulating layer **1610**. For example, the second bonding sheet **1630** may include a first bonding part, a second bonding part, and an extension bonding part between the first bonding part and the second bonding part.

[0259] The first bonding part of the second bonding sheet **1630** may be disposed between the first insulating part and the first lead pattern part.

[0260] The second bonding part of the second bonding sheet **1630** may be disposed between the second insulating part and the second lead pattern part.

[0261] In addition, the extension bonding part of the second bonding sheet **1630** may be disposed between the extension insulating part and the extension pattern part.

[0262] The image sensor substrate according to the embodiment provides elastic force in the extension insulating part of the insulating layer **1610**, the extension bonding part of the second bonding sheet **1630**, and the extension pattern part of the conductive pattern portion **1640**, and the image sensor **1700** can be moved. Preferably, the extension insulating part, the extension bonding part, and the extension pattern part may also be referred to as an elastic member. In addition, the elastic member may have a layered structure including an insulating layer region, a bonding layer region, and a pattern layer region. This will be described in more detail below.

[0263] In addition, the second bonding sheet **1630** may be omitted for the image sensor substrate according to the embodiment. For example, the conductive pattern portion **1640** may be directly disposed on the insulating layer **1610** without the second bonding sheet **1630**. That is, the upper surface of the extension insulating part and the lower surface of the extension pattern part may directly contact each other without the extension bonding part of the insulating layer **1610**.

Accordingly, in an embodiment, only the extension insulating part and the extension pattern part may serve as the elastic member without the extension bonding part.

[0264] The supporting layer **1650** may be a support substrate. A third driving part **1530** may be mounted on the supporting layer **1650**, and accordingly, an electric signal may be applied to the third driving part **1530**.

[0265] In addition, the third driving part **1530** may include an electromagnet, and thus may be mounted on the housing.

[0266] The supporting layer **1650** may support the insulating layer **1610**. Preferably, the supporting layer **1650** may support the first bonding sheet **1620**, the second bonding sheet **1630**, the insulating layer **1610**, the conductive pattern portion **1640** and the image sensor **1700** at a position floating from the bottom surface of the housing **1300**.

[0267] The image sensor **1700** is disposed on the image sensor substrate **1600**,

[0268] One end of the flexible circuit board **1800** may be connected to the image sensor substrate **1600**.

[0269] To this end, a part of a region of the flexible circuit board **1800** is disposed inside the

housing **1300**, and accordingly, may be connected to the conductive pattern portion **1640** of the image sensor substrate **1600**.

[0270] Hereinafter, the substrate **1600** for an image sensor according to the second embodiment will be described in more detail.

[0271] FIG. **18** is a view showing the insulating layer shown in FIG. **17**.

[0272] Referring to FIG. **18**, the insulating layer **1610** may elastically support the conductive pattern portion **1640** constituting the image sensor substrate **1600**.

[0273] The insulating layer **1610** may have an elastic region having a certain elastic force to move the image sensor **1700** disposed on the image sensor substrate **1600** in the X-axis, Y-axis, and Z-axis directions.

[0274] To this end, the elastic region may include the extension insulating part **1614** having an elastic force of the insulating layer **1610**. Specifically, the first insulating part **1611** may be disposed in the center of the insulating layer **1610**. In addition, the second insulating part **1612** may be disposed around the first insulating part **1611** at a position spaced apart from the first insulating part **1611** by a predetermined distance.

[0275] In addition, the insulating layer **1610** may include the first insulating part **1611** and the second insulating part **1612**, and an open region can be between the include the first insulating part **1611** and the second insulating part **1612**. Preferably, the first insulating part **1611** and the second insulating part **1612** may be spaced apart from each other by the open region **1613** interposed therebetween.

[0276] In addition, the extension insulating part **1614** may have one end connected to the first insulating part **1611** and the other end connected to the second insulating part **1612**. To this end, the extension insulating part **1614** may be formed in plural. Preferably, the extension insulating part **1614** may include first to fourth extension insulating parts **1614**.

[0277] The first insulating part **1611** may include a plurality of outer portions. Preferably, the first insulating part **1611** may include a left outer portion, a right outer portion, an upper outer portion, and a lower outer portion.

[0278] The second insulating part **1612** may include a plurality of inner parts. Preferably, the second insulating part **1612** may include a left inner portion, a right inner portion, an upper inner portion, and a lower inner portion.

[0279] In this case, the left outer portion of the first insulating part **1611** and the left inner portion of the second insulating part **1612** may be disposed to face each other. In addition, the first extension insulating part constituting the extension insulating part **1614** may connect the left outer portion of the first insulating part **1611** and the left inner portion of the second insulating part **1612** to each other.

[0280] In addition, the right outer portion of the first insulating part **1611** and the right inner portion of the second insulating part **1612** may be disposed to face each other. In addition, the second extension insulating part constituting the extension insulating part **1614** may connect the right outer portion of the first insulating part **1611** and the right inner portion of the second insulating part **1612** to each other.

[0281] In addition, the upper outer portion of the first insulating part **1611** and the upper inner portion of the second insulating part **1612** may be disposed to face each other. In addition, the third extension insulating part constituting the extension insulating part **1614** may connect the upper outer portion of the first insulating part **1611** and the upper inner portion of the second insulating part **1612** to each other.

[0282] In addition, the lower outer portion of the first insulating part **1611** and the lower inner portion of the second insulating part **1612** may be disposed to face each other. In addition, the fourth extension insulating part constituting the extension insulating part **1614** may connect the lower outer portion of the first insulating part **1611** and the lower inner portion of the second insulating part **1612** to each other.

[0283] In addition, the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be integrally formed. Accordingly, it is possible to further utilize the elastic force of the insulating layer **1610** when the image sensor is tilted, and it is possible to inhibit detachment between the first insulating part **1611**, the extension insulating part **1614**, and the second insulating part **1612**.

[0284] That is, the insulating layer **1610** may be etched or physically punched to have a spring shape of the extension insulating part **1614** on one insulating member to form the open region **1613**. Accordingly, the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be formed of the same insulating material. However, the embodiment is not limited thereto.

[0285] In other words, each of the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be formed in separate configurations. That is, the insulating layer **1610** according to the embodiment may be formed by further forming a configuration corresponding to the extension insulating part **1614** between the first insulating part **1611** and the second insulating part **1612**, after preparing the first insulating part **1611** and the second insulating part **1614**.

[0286] On the other hand, the length of the extension insulating part **1614** is set to have at least 1.5 times the linear distance between the first insulating part **1611** and the second insulating part **1612**. In addition, the length of the extension insulating part **1614** is set to be 20 times or less than the linear distance between the first insulating part **1611** and the second insulating part **1614**. Preferably, the length of the extension insulating part **1614** is set to be less than 4 times the linear distance between the first insulating part **1611** and the second insulating part **1612**.

[0287] Here, the linear distance may mean a distance between an outer surface and an inner surface facing each other in the first insulating part **1611** and the second insulating part **1612**. Preferably, the linear distance may be a distance between the left outer surface of the first insulating part **1611** and the left inner surface of the second insulating part **1612**. In addition, the linear distance may be a distance between the right outer surface of the first insulating part **1611** and the right inner surface of the second insulating part **1612**. Also, the linear distance may be a distance between an upper outer surface of the first insulating part **1611** and an upper inner surface of the second insulating part **1612**. Also, the linear distance may be a distance between a lower outer surface of the first insulating part **1611** and a lower inner surface of the second insulating part **1612**.

[0288] Meanwhile, a linear distance between the first insulating part **1611** and the second insulating part **1612** may be 1.5 mm. At this time, if less than 1.5 times the linear distance between the first insulating part **1614** and the second insulating part **1612** of the extension insulating part **1614**, the mobility of the image sensor substrate **1600** may be reduced due to the decrease in elastic force of the extension insulating part **1614**. In addition, if the length of the extension insulating part **1614** is greater than 20 times the linear distance between the first insulating part **1614** and the second insulating part **1612**, an image sensor **1700** disposed on the insulating layer **1610** cannot be stably supported, and thus a problem may occur in movement accuracy. Accordingly, in the embodiment, in order to improve mobility, the length of the extension insulating part **1614** is set to be less than 4 times the linear distance between the first insulating part **1611** and the second insulating part **1612**.

[0289] Accordingly, the extension insulating part **1614** may be formed to have a plurality of bent spring shapes in the open region **1613**.

[0290] Meanwhile, at least one slit (not shown) may be formed in the first insulating part **1611** of the insulating layer **1610**. The slit may be formed to maintain flatness of the insulating layer **1610**. That is, the conductive pattern portion **1640** and the image sensor **1700** are disposed on the insulating layer **1610**. In addition, a first driving part is disposed under the insulating layer **1610**. In this case, the flatness of the first driving part or the image sensor **1700** directly affects the reliability of the camera module, and image quality may deteriorate as the flatness deteriorates. Accordingly, in the embodiment, by forming at least one slit in the insulating layer **1610**, not only the weight of

the insulating layer **1610** is reduced, but also the flatness can be maintained, thereby improving the overall reliability of the camera module.

[0291] FIG. **19** is a view showing a modified example of the conductive pattern portion shown in FIG. **17**.

[0292] Referring to FIG. **19**, the insulating layer **1610A** may include an insulating part **1611** and a plurality of open holes **1613A**, **1613B**, **1613C**, and **1613D** disposed in the insulating part **1611**.

[0293] That is, in the first embodiment, the open region has a shape of one closed loop that is entirely disposed around the first insulating part. In contrast, in the second embodiment, a plurality of open holes **1613A**, **1613B**, **1613C**, and **1613D** are disposed at positions spaced apart from each other by a predetermined interval, so that the first and second insulating parts of the insulating layer **1610A** are not separated from each other. In this case, the previously described extension insulating part **1614** may be disposed in the plurality of open holes **1613A**, **1613B**, **1613C**, and **1613D**.

[0294] In conclusion, in the modified example, the corner regions between the first and second insulating parts are interconnected, and open regions are formed only in the remaining regions excluding the corner regions. In addition, the extension insulating part **1614** is disposed in the open region. Accordingly, in a modified example, while improving the support force of the image sensor **1700** due to the increase in the strength of the insulating layer **1610**, the reliability of mobility by the extension insulating part **1614** may be improved.

[0295] FIG. **20** is a view showing a second bonding sheet shown in FIG. **17**.

[0296] Referring to FIG. **20**, the second bonding sheet **1630** is disposed on the insulating layer **1610**. In this case, the second bonding sheet **1630** has a planar shape corresponding to the planar shape of the insulating layer **1610**.

[0297] That is, the second bonding sheet **1630** may include a first bonding part **1631** disposed on the first insulating part **1611** of the insulating layer and a second bonding part **1632** disposed on the second insulating part **1612** of the insulating layer **1610**. In addition, the second bonding sheet **1630** may include an open region **1633** between the first bonding part **1631** and the second bonding part **1632**.

[0298] The second bonding part **1632** is disposed to surround the first bonding part **1631** at a position spaced apart from the first bonding part **1631** by a predetermined distance. In this case, the second bonding part **1632** does not directly contact the first bonding part **1631**. Accordingly, the first bonding part **1631** and the second bonding part **1632** may be separated from each other through the open region **1633**.

[0299] In addition, the extension bonding part **1634** connecting between the first bonding part **1631** and the second bonding part **1632** is disposed in the open region **1633**. The extension bonding part **1634** has a shape corresponding to the extension insulating part **1614**. The extension bonding part **1634** may be disposed in a region vertically overlapping the extension insulating part **1614**. In this case, the extension bonding part **1634** may have a plane area corresponding to the extension insulating part **1614**. Preferably, the plane area of the extension bonding part **1634** may be 0.9 to 1.1 times the plane area of the extension insulating part **1614**. The extension bonding part **1634** of the second bonding sheet **1630** may be formed through the same process as the extension insulating part **1614** of the insulating layer **1610**. Accordingly, the extension bonding part **1634** may have the same shape as the extension insulating part **1614** and have the same plane area.

[0300] The second bonding sheet **1630** may be formed of a double-sided adhesive film. The second bonding sheet **1630** may be formed of an epoxy or acrylic adhesive or a thermosetting adhesive film.

[0301] The second bonding sheet **1630** may have a thickness of 25 μm .

[0302] Meanwhile, a first bonding sheet **1610** is disposed under the insulating layer **1610**. In this case, the first bonding sheet **1610** is disposed under the second insulating part **1612** of the insulating layer **1610**. In other words, the first bonding sheet **1610** may be selectively formed only in a region of the lower surface of the insulating layer **1610** in which the supporting layer **1650** is

disposed. Meanwhile, the supporting layer **1650** and the insulating layer **1610** may be integrally formed with each other. Accordingly, the first bonding sheet **1610** may be selectively omitted. [0303] That is, the supporting layer **1650** may be formed by removing a portion of the lower surface of the insulating member constituting the insulating layer **1610** and the supporting layer **1650** to form a groove.

[0304] FIG. **21** is a view showing a conductive pattern portion shown in FIG. **17**.

[0305] Referring to FIG. **21**, the conductive pattern portion **1640** may be disposed on the insulating layer **1610** with a specific pattern. The conductive pattern portion **1640** includes a first conductive pattern part **1641** disposed on a first region of the insulating layer **1610** and a second conductive pattern part **1642** disposed on a second region of the insulating layer **1610**, a third conductive pattern part **1643** disposed on a third region of the insulating layer **1610**, and a fourth conductive pattern part **1644** disposed on a fourth region of the insulating layer **1610**. In the drawing, a region vertically overlapping the first insulating part **1611** of the insulating layer **1610** and the first bonding part **1631** of the second bonding sheet **1630** is referred to as 'A', a region vertically overlapping the second insulating part **1612** of the insulating layer **1610** and the second bonding part **1632** of the second bonding sheet **1630** is referred to as 'B', and a region vertically overlapping the open region **1613** of the insulating layer **1610** and the open region **1633** of the second bonding sheet **1630** is referred to as 'C'.

[0306] The first conductive pattern part **1641** may be disposed on a left side of an upper surface of the insulating layer **1610**. That is, the first conductive pattern part **1641** may be disposed on a left region of the first insulating part **1611**, a left region of the second insulating part **1612**, and the first extension insulating part **1614** connecting these. Preferably, the first conductive pattern part **6410** may be disposed on a part of the second bonding sheet on the left region of the first insulating part **1611**, a part of the second bonding sheet on the left area of the second insulating part **1612**, and a part of the extension bonding part connecting them to each other.

[0307] In addition, the first conductive pattern part **1641** may be disposed on the insulating layer **1610** without the second bonding sheet **1630**.

[0308] The second conductive pattern part **1642** may be disposed in a right region of an upper surface of the insulating layer **1610**. That is, the second conductive pattern part **1642** may be disposed on a right region of the first insulating part **1611**, a right region of the second insulating part **1612** and a second extension insulating part connecting these.

[0309] The third conductive pattern part **1643** may be disposed in an upper region of the upper surface of the insulating layer **1610**. That is, the third conductive pattern part **1643** may be disposed on an upper region of the first insulating part **1611**, an upper region of the second insulating part **1612**, and a third extension insulating part connecting these.

[0310] The fourth conductive pattern part **1644** may be disposed in a lower region of the upper surface of the insulating layer **1610**. That is, the fourth conductive pattern part **1644** may be disposed on a lower region of the first insulating part **1611**, a lower region of the second insulating part **1612**, and a fourth extension insulating part connecting these.

[0311] As described above, the conductive pattern portions **1640** are respectively disposed on different regions, and accordingly, the elastic support force in the movement of the image sensor substrate **1600** may be increased. That is, if the conductive pattern portion **1640** is intensively disposed only in a specific area, reliability in moving the image sensor substrate **1600** in a specific direction may be lowered. For example, if the conductive pattern portion **1640** includes only the first and second conductive pattern portions, there is no problem with the movement of the image sensor substrate **1600** in the X-axis direction, stability may be deteriorated when the image sensor substrate **1600** moves in the Y-axis direction. In addition, in this case, the conductive pattern portion **1640** may be disconnected according to the periodic movement of the image sensor substrate **1600**. Accordingly, in the embodiment, the conductive pattern portion **1640** is distributed in each of the four regions as described above, so that the image sensor substrate **1600** can be

stably moved in the X-axis, Y-axis, and Z-axis. That is, the conductive pattern portion may be formed to be symmetrical up/down/left/right on the insulating layer **1610**.

[0312] Meanwhile, the conductive pattern portion **1640** may include a first lead pattern part **1645** connected to the image sensor **1700** and a second lead pattern part **1646** connected to the flexible circuit board **1800**. In addition, the conductive pattern portion **1640** may include an extension pattern part **1647** connecting between the first lead pattern part **1645** and the second lead pattern part **1646**.

[0313] The first lead pattern part **1645** is disposed on the first insulating part **1611** of the insulating layer **1610**. The first lead pattern part **1645** may be disposed on an outer region of the first insulating part **1611**. That is, the image sensor mounting area on which the image sensor **1700** is mounted may be included on the first insulating part **1611**. In this case, the image sensor mounting area may be a central area of the first insulating part **1611**. Accordingly, the first lead pattern part **1645** may be disposed around the image sensor mounting area of the first insulating part **1611**.

[0314] The second lead pattern part **1646** is disposed on the second insulating part **1612** of the insulating layer **1610**. The second lead pattern parts **1646** may be disposed on the second insulating part **1612**, respectively. In this case, the first lead pattern part **1645** and the second lead pattern part **1646** may be disposed to face each other on the first insulating part **1611** and the second insulating part **1612**. That is, the number of first lead pattern parts **1645** may be the same as the number of second lead pattern parts **1646**. In addition, each of the first lead pattern parts **1645** may be disposed to face the second lead pattern part **1646**, respectively.

[0315] Meanwhile, the extension pattern part **1647** may be disposed between the first lead pattern part **1645** and the second lead pattern part **1646**.

[0316] The extension pattern part **1647** may have one end connected to the first lead pattern part **1645** and the other end connected to the second lead pattern part **1646** facing the first lead pattern part **1645**.

[0317] In this case, the extension pattern part **1647** may be disposed on the extension insulating part **1614** of the insulating layer **1610**. Preferably, the extension pattern part **1647** may be disposed on the extension bonding part **1634** disposed on the extension insulating part **1614**. Preferably, the extension bonding part **1634** may be disposed between the extension insulating part **1614** and the extension pattern part **1647**. In addition, the upper surface of the extension bonding part **1634** may directly contact the lower surface of the extension pattern part **1647**. In addition, the lower surface of the extension bonding part **1634** may directly contact the extension insulating part **1614**.

Preferably, the extension pattern part **1647**, the extension insulating part **1614**, and the extension bonding part **1634** may be overlapped and aligned in a mutually vertical direction.

[0318] The length of the extension pattern part **1647** is greater than a linear distance between the first lead pattern part **1645** and the second lead pattern part **1646**. That is, the extension pattern part **1647** may be formed to have a structure that is bent a plurality of times between the first lead pattern part **1645** and the second lead pattern part **1646**. Preferably, the extension pattern part **1647** may be formed to have a spring shape between the first lead pattern part **1645** and the second lead pattern part **1646**.

[0319] At this time, the extension pattern part **1647** may be formed by etching to have the above shape through an additive process, a subtractive process, a modified semi-additive process (MSAP), a semi-additive process (SAP), and the like. Preferably, the extension pattern part **1647** may be formed simultaneously with the first lead pattern part **1645** and the second lead pattern part **656**. More preferably, the extension pattern part **1647** may be integrally formed with the first lead pattern part **1645** and the second lead pattern part **1646**.

[0320] Meanwhile, the thickness of the conductive pattern portion **1640** including the extension pattern part **1647**, the first lead pattern part **1645**, and the second lead pattern part **1646** may be 10 μm to 50 μm . For example, the thickness of the conductive pattern portion **1640** may be 30 μm to 40 μm . In this case, when the thickness of the conductive pattern portion **1640** is less than 10 μm ,

the conductive pattern portion **1640** may be broken when the image sensor substrate **1600** moves. In addition, when the thickness of the conductive pattern portion **1640** is greater than 50 μm , elastic force of the extension pattern part **1647** may be lowered, thereby hindering the mobility of the image sensor substrate **1600**. Accordingly, in the embodiment, the thickness of the conductive pattern portion **1640** is set to be $35\ \mu\text{m} \pm 5\ \mu\text{m}$ such that the image sensor substrate **1600** may be stably moved.

[0321] In addition, the length of the extension pattern part **1647** is set to have at least 1.5 times the linear distance between the first lead pattern part **1645** and the second lead pattern part **1646**. In addition, the length of the extension pattern part **1647** is set to be 20 times or less of the linear distance between the first lead pattern part **1645** and the second lead pattern part **1646**. Preferably, the length of the extension pattern part **1647** is set to be less than or equal to 4 times the linear distance between the first lead pattern part **1645** and the second lead pattern part **1646**.

[0322] The linear distance between the first lead pattern part **1645** and the second lead pattern part **1646** may be 1.5 mm.

[0323] In this case, when the length of the extension pattern part **1647** is smaller than 1.5 times the linear distance between the first lead pattern part **1645** and the second lead pattern part, the mobility of the image sensor substrate **1600** may be reduced due to the decrease in the elastic force of the extension pattern part **1647**. In addition, when the length of the extension pattern part **1647** is greater than 20 times the linear distance, a resistance increases as a signal transmission distance is increased by the extension pattern part **1647**, and accordingly, noise may be included in a signal transmitted via the extension pattern part **1647**. Accordingly, in order to minimize noise generation, the length of the extension pattern part **1647** is set to be 4 times or less the linear distance between the first lead pattern part **1645** and the second lead pattern part **164**.

[0324] Meanwhile, the line width of the extension pattern part **1647** may be smaller than the line width of the extension insulating part **1614**. Accordingly, the extension insulating part **1614** may include a first region overlapping with the extension pattern part **1647** in a vertical direction, and a second region excluding the first region. In this case, when the line width of the extension pattern part **1647** is smaller than the line width of the extension insulating part **1614**, the lower surface of the extension pattern part **1647** may be exposed as the lower surface of the insulating layer **1610**. In addition, due to such exposure, the extension pattern part **1647** may contact other components while the image sensor **1700** is moving, and thus reliability may be problematic. Accordingly, in the embodiment, the line width of the extension insulating part **1614** is larger than the line width of the extension pattern part **1647** disposed thereon.

[0325] FIG. 22 is a plan view of a substrate for an image sensor according to a second embodiment.

[0326] Referring to FIG. 22, the conductive pattern portion **1640** is disposed on the insulating layer **1610**. In this case, an open region is formed in each of the insulating layers **1610**. In addition, the second bonding sheet **1630** may be disposed on the insulating layer **1610**. In addition, the conductive pattern portion **1640** is disposed on the second bonding sheet **1630**. In this case, open regions are formed in each of the insulating layer **1610** and the second bonding sheet **1630**, and the open regions may overlap each other in a vertical direction.

[0327] In addition, an elastic member having an elastic force may be disposed in each of the open regions. Here, the elastic member may be a part of the insulating layer **1610**, or a part of the insulating layer **1610** and a part of the second bonding sheet **1630**. That is, the elastic member may be formed as a part of the insulating layer **1610** or/and the second bonding sheet **1630**. That is, the extension insulating part **1614** having a bent shape that is bent a plurality of times is disposed in the open region of the insulating layer **1610**. The extension insulating part **1614** may have a spring shape. In addition, an extension bonding part **1634** having a bent shape that is bent a plurality of times is disposed in the open region of the second bonding sheet **1630**. The extension bonding part **1634** may have a spring shape.

[0328] In addition, the extension bonding part **1634** and the extension insulating part **1614** may have the same shape while having the same planar area.

[0329] Meanwhile, each of line widths of the extension bonding part **1634** and the extension insulating part **1614** may be larger than the line width of the extension pattern part **1647**.

Accordingly, when the image sensor substrate is viewed from above, at least a part of the extension bonding part **1634** and the extension insulating part **1614** positioned under the extension pattern part **1647** may be exposed.

[0330] At this time, the image sensor substrate **1600** may move not only in the X-axis direction and the Y-axis direction, but also in the Z-axis direction. At this time, there is a difference between the elastic modulus of the extension bonding part **1634** and the extension insulating part **1614** and the elastic modulus of the extension pattern part **1647**. Accordingly, when the image sensor substrate **1600** moves in the Z-axis direction, the movement distances of the extension bonding part **1634** and the extension insulating part **1614** may appear differently. In addition, this causes a situation in which the extension pattern part **1647** is in contact with a constituent portion of another metal material, thereby causing an electrical reliability problem (e.g., a short). Accordingly, in the embodiment, the line widths of the extension insulating part **1614** and the extension insulating part **1614** are made larger than the line width of the extension pattern part **1647** as described above, so that the electrical reliability problem can be solved.

[0331] FIG. **23** is a view showing a connection structure between a flexible circuit board and an image sensor board according to a second embodiment.

[0332] Referring to FIG. **23**, the flexible circuit board **1800** electrically connects the image sensor board **1600** and an external main board (not shown) to each other.

[0333] One end of the flexible circuit board **1800** may be connected to the image sensor substrate **1600**. The flexible circuit board **1800** may receive an electrical signal output from the image sensor **1700**. The flexible circuit board **1800** may include a connector **1810** at the other end. A main board (not shown) may be connected to the connector **1810**.

[0334] That is, the flexible circuit board **1800** may connect the camera module and the main board of the external device. Specifically, the flexible circuit board **1800** may connect between the conductive pattern portion **1650** of the image sensor substrate **1600** of the camera module and the main board of a portable terminal.

[0335] To this end, an area of the flexible circuit board **1800** is disposed inside the housing **1300**, and accordingly, may be connected to the conductive pattern portion **1650** of the image sensor substrate **1600**.

[0336] That is, the flexible circuit board **1800** may include a first connector part **1801**, a second connector part **1803**, and a connection part **1802**.

[0337] The first connector part **1801** may be disposed inside the housing **1300**. The first connector part **1801** may include a plurality of pads **1804**, **1805**, **1806**, and **1807** connected to the conductive pattern portion **1640**.

[0338] The first connector part **1801** may be electrically connected to the second lead pattern part **1646** of the conductive pattern portion **1640**. That is, the plurality of pad portions **1804**, **1805**, **1806**, and **1807** of the first connector part **1801** may be electrically connected to the second lead pattern part **1646**.

[0339] To this end, the first connector part **1801** may include a first pad part **1804** connected to the second lead pattern part **1646** of the first conductive pattern part **1641**. In addition, the first connector part **1801** may include a second pad part **1805** connected to the second lead pattern part **1646** of the second conductive pattern part **1642**. Further, the first connector part **1801** may include a third pad part **1806** connected to the second lead pattern part **1646** of the third conductive pattern part **1643**. Further, the first connector part **1801** may include a fourth pad part **1807** connected to the second lead pattern part **1646** of the fourth conductive pattern part **1644**.

[0340] In this case, the first connector part **1801** has a shape corresponding to the second insulating

part **1612** of the insulating layer **1610**, and it may be disposed surrounding the upper region of the second lead pattern part **1646** of the conductive pattern portion **1640**, and the plurality of pad portions **1804**, **1805**, **1806**, and **1807** may be disposed on a lower surface thereof.

[0341] The connection part **1802** connects the first connector part **1801** and the second connector part **1803**. A part of the connection part **1802** may be disposed in the housing **1300**, and may be extended therefrom to be exposed to the outside of the housing **1300**.

[0342] The second connector part **1803** may include a connector **1810** connected to the main board of the terminal.

[0343] Meanwhile, the image sensor **1700** may be attached on the first insulating part **1631** of the insulating layer **1610**. In this case, the image sensor **1700** may be attached on the first insulating part **1611** so that the electrode **710** faces upward.

[0344] In addition, a connection member **1720**, such as a metal wire, is formed between the electrode **710** of the image sensor **1700** and the first lead pattern part **1645**, so that the connection member **1720** may electrically connect the electrode of the image sensor and the first lead pattern part.

[0345] FIG. **24** is a diagram specifically illustrating a layer structure of a conductive pattern portion according to a second exemplary embodiment.

[0346] Referring to FIG. **24**, the conductive pattern portion **1640** is disposed on the second bonding sheet **1630** disposed on the insulating layer **1610**. In this case, the conductive pattern portion **1640** includes a first lead pattern part **1645** disposed on the first insulating part **1611** of the insulating layer **1610**, a second lead pattern part **1646** disposed on the second insulating part **1612**, and an extension pattern part **1647** connecting therebetween.

[0347] In this case, each of the first lead pattern part **1645**, the second lead pattern part **1646**, and the extension pattern part **1647** may include a metal layer **1640A** and a plating layer **1640B**.

[0348] The metal layer **1640A** may be disposed on the second bonding sheet **1630**. That is, the metal layer **1640A** is disposed on the second bonding sheet **1630** to configure the first lead pattern part **1645**, the second lead pattern part **1646**, and the extension pattern part **1647**, respectively

[0349] The plating layer **1640B** may be disposed on the metal layer **1640A**. Preferably, the plating layer **1640B** may be a surface treatment layer disposed on the metal layer **1640A**.

[0350] The plating layer **1640B** includes any one of Ni/Au alloy, gold (Au), electroless nickel immersion gold (ENIG), Ni/Pd alloy, and organic compound plating (Organic Solderability Preservative, OSP).

[0351] In this case, the plating layers **1640B** constituting the first lead pattern part **1645** and the second lead pattern part **1646** may correspond to each other. Alternatively, the plating layer **1640B** constituting the extension pattern part **1647** may have the same thickness as the plated layer **650B** constituting the first lead pattern part **1645** and the second lead pattern part **1646**.

[0352] That is, the plating layer **1640B** of each of the first lead pattern part **1645**, the second lead pattern part **1646** and the extension pattern part **1647** may be formed on the upper and side surfaces the corresponding metal layer **1640A**.

[0353] Meanwhile, the thickness of the plating layer **1640B** may be 0.3 μm to 1 μm . For example, the thickness of the plating layer **1640B** may be 0.3 μm to 0.7 μm . The thickness of the plating layer **1640B** may be 0.3 μm to 0.5 μm .

[0354] FIG. **25** is a view showing a camera device according to a third embodiment. Prior to the description of FIG. **25**, components that are substantially the same as those of FIG. **17** are denoted by the same reference numerals.

[0355] Referring to FIG. **25**, the camera device may include a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter part **1400**, a driving part **1510**, **1520**, **1530**, an image sensor substrate **1600**, an image sensor **1700**, and a flexible circuit board **1800**. Here, a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter part **1400**, a driving part **1510**, **1520**, **1530**, an image sensor **1700**, and a flexible circuit board **1800** are substantially the same as

the structure of FIG. 17, and thus a detailed description thereof will be omitted.

[0356] The image sensor substrate **1600** in FIG. 25 may include an insulating layer **1610**, a first bonding sheet **1620**, a conductive pattern portion **1640**, and a supporting layer **1650**. That is, compared with the image sensor substrate of FIG. 17, in the image sensor substrate of FIG. 25, the second bonding sheet has been removed on the insulating layer **1610**, and accordingly, the conductive pattern portion **1640** may be disposed in direct contact with the insulating layer **1610**. In this case, a bonding layer (not shown) may be selectively disposed on the insulating layer **1610** only in a region where the image sensor **1700** is disposed.

[0357] The structures of the insulating layer **1610** and the conductive pattern portion **1640** in FIG. 25 are the same as those of the insulating layer and the conductive pattern portion **1640** described with reference to FIG. 17. Accordingly, the upper surface of the first insulating part **1611** may directly contact the lower surface of the first lead pattern part **1645**. In addition, the upper surface of the second insulating part **1612** may directly contact the lower surface of the second lead pattern part **1646**. In addition, the upper surface of the extension insulating part **1614** may directly contact the lower surface of the extension pattern part **1647**.

[0358] FIG. 26A is a view showing a camera device according to a fourth embodiment, and FIG. 26B is a plan view of a second bonding sheet in FIG. 26A. Prior to the description of FIGS. 26A and 26B, the same reference numerals are assigned to substantially the same components as those of FIG. 17.

[0359] FIG. 26A and FIG. 26B, the camera device may include a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter part **1400**, a driving part **1510**, **1520**, and **1530**, and an image sensor substrate **1600**, an image sensor **1700**, and a flexible circuit board **1800**. Here, a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter part **1400**, a driving part **1510**, **1520**, **1530**, an image sensor **1700**, and a flexible circuit board **1800** are substantially the same as the structure of FIG. 17, and thus a detailed description thereof will be omitted.

[0360] The image sensor substrate **1600** in FIGS. 26A and 26B may include an insulating layer **1610**, a first bonding sheet **1620**, a second bonding sheet **1630**, a conductive pattern part **1640**, and a support layer **1650**. That is, compared with the image sensor substrate of FIG. 17, the second bonding sheet **1630** of the image sensor substrate of FIG. 26A may be disposed only in a partial area on the insulating layer **1610**. That is, the second bonding sheet **1630** may be selectively disposed only in a region of the upper surface of the insulating layer **1610** in which the conductive pattern portion **1640** is to be disposed.

[0361] That is, referring to FIG. 26B, the second bonding sheet **1630** may include a first bonding part **1631**, a second bonding part **1632**, and an extension bonding part **1634**.

[0362] The first bonding part **1631** may be disposed between the first insulating part **1611** and the first lead pattern part **1645**. The first bonding part **1631** may have a shape corresponding to the first lead pattern part **1645**. Preferably, the first bonding part **1631** may have a plane area equal to that of the first lead pattern part **1645**.

[0363] The second bonding part **1632** may be disposed between the second insulating part **1612** and the second lead pattern part **1646**. The second bonding part **1632** may have a shape corresponding to the second lead pattern part **1646**. Preferably, the second bonding part **1632** may have the same plane area as the plane area of the second lead pattern part **1646**.

[0364] The extension bonding part **1634** may be disposed between the extension insulating part **1614** and the extension pattern part **1647**. The extension bonding part **1634** may have a shape corresponding to the extension pattern part **1647**. Preferably, the extension bonding part **1634** may have the same plane area as that of the extension pattern part **1647**.

[0365] Meanwhile, the second bonding sheet **1630** in the embodiment may be a seed layer of the conductive pattern portion **1640**. Accordingly, when the conductive pattern portion **1640** is patterned, the second bonding sheet **1630** may also be removed together with the conductive pattern portion **1640**, and accordingly, the second bonding sheet **1630** may have the same shape as

the conductive pattern portion **1640**.

[0366] According to an embodiment, in order to implement the OIS and AF functions of the camera module, instead of moving the conventional lens barrel, the image sensor is moved relative to the lens barrel in the X-axis, Y-axis and Z-axis directions. Accordingly, the camera module according to the embodiment may remove a complex spring structure for implementing the OIS and AF functions, and thus the structure may be simplified. In addition, by moving the image sensor according to the embodiment relative to the lens barrel, it is possible to provide a stable structure compared to the conventional structure.

[0367] In addition, according to an embodiment, the extension pattern part electrically connected to the image sensor has a spring structure and is disposed in a floating form on the spring plate.

[0368] In addition, in the insulating layer, an extension insulating part having a spring shape is disposed in a region vertically overlapping with the extension pattern part. Accordingly, the camera module may elastically support the image sensor more stably and move the image sensor with respect to the lens barrel.

[0369] In addition, the length of the extension pattern part in the embodiment is set to be at least 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part. Accordingly, noise generation can be minimized while improving the mobility of the image sensor substrate.

[0370] In addition, in the embodiment, the width of the extension insulating part is made larger than the width of the extension pattern part, so that the extension pattern part can be stably supported by the extension insulating part, thereby improving operational reliability.

[0371] In addition, according to the embodiment, the elastic member and the extension pattern part are not aligned with each other in a vertical direction, thereby solving an electrical reliability problem that may occur due to contact between the elastic member and the extension pattern part.

Claims

1. A substrate for an image sensor comprising: a spring plate; an insulating layer disposed on the spring plate and including a first open region; and a conductive pattern part disposed on the insulating layer, wherein the insulating layer includes a first insulating part, a second insulating part spaced apart from the first insulating part with the first open region therebetween, and a third insulating part disposed in the first open region and connecting between the first insulating part and the second insulating part, wherein the conductive pattern part includes an extension pattern part disposed on the third insulating part, wherein the spring plate includes an elastic member disposed under the third insulating part, and wherein the third insulating part includes a first portion overlapping the elastic member in a thickness direction of the insulating layer, and a second portion connected to the first portion and not overlapping the elastic member in the thickness direction.
2. The substrate of claim 1, wherein the spring plate includes first and second plate parts spaced apart from each other with a second open region therebetween, wherein the elastic member connects the first and second plate parts, wherein the first insulating part is disposed on the first plate part, wherein the second insulating part is disposed on the second plate part, and wherein the conductive pattern part includes a first lead pattern part disposed on the first insulating part, and a second lead pattern part disposed on the second insulating part, and wherein the extension pattern part connects between the first lead pattern part and the second lead pattern part, and includes a bent part.
3. The substrate of claim 2, wherein the extension pattern part includes a plurality of extension patterns on the first open region and spaced apart from each other, and wherein the third insulating part includes a plurality of extension insulating parts respectively corresponding to each of the plurality of extension patterns and formed in the first open region.
4. The substrate of claim 3, wherein the second open region overlaps the first open region in the

thickness direction.

5. The substrate of claim 3, wherein the elastic member includes a bent part disposed in a corner region of the second open region adjacent to a corner region of the first plate part and bent in different horizontal directions.

6. The substrate of claim 5, wherein the second open region includes a plurality of corner regions adjacent to different corner regions of the first plate part, wherein the elastic member includes a plurality of bent parts disposed in each of the plurality of corner regions of the second open region, wherein the extension patterns include a first extension pattern and a second extension pattern adjacent to the first extension pattern, wherein the plurality of extension insulating parts include a first extension insulating part overlapping the first extension pattern in the thickness direction and a second extension insulating part overlapping the second extension pattern in the thickness direction, wherein the first extension insulating part does not overlap the second extension pattern in the thickness direction, and wherein the second extension insulating part does not overlap the first extension pattern in the thickness direction.

7. The substrate of claim 2, wherein a length of the extension pattern part is greater than a straight-line distance between the first lead pattern part and the second lead pattern part.

8. The substrate of claim 7, wherein the length of the extension pattern part is 1.5 to 4 times the straight-line distance between the first lead pattern part and the second lead pattern part.

9. The substrate of claim 6, wherein each of the first and second extension insulating parts has a line width larger than a line width of each of the first and second extension patterns.

10. The substrate of claim 2, wherein each of the first lead pattern part and the second lead pattern part includes a metal layer, and a plating layer disposed on the metal layer.

11. The substrate of claim 1, wherein the elastic member has a region in the first open region that does not overlap with the extension pattern part in the thickness direction.

12. The substrate of claim 1, wherein the extension pattern part includes a binary alloy of any one of copper (Cu)-nickel (Ni), copper (Cu)-tin (Sn), copper (Cu)-beryllium (Be), or copper (Cu)-cobalt (Co), or a ternary alloy of any one of copper (Cu)-nickel (Ni)-tin (Sn) or copper (Cu)-beryllium (Be)-cobalt (Co).

13. The substrate of claim 2, wherein the first plate part includes at least one slit.

14. The substrate of claim 3, wherein the extension pattern part has a spring shape including a plurality of bent portions, and wherein the plurality of extension insulating parts have a shape corresponding to the extension pattern portion.

15. The substrate of claim 2, wherein a line width of the extension pattern part is smaller than a line width of each of the first and second lead pattern parts.

16. A camera module comprising: a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; a substrate for an image sensor according to claim 1, which is disposed in the housing; an image sensor disposed on a first lead pattern part of the substrate for the image sensor; and a flexible circuit board including a pad part connected to a second lead pattern part of the substrate for the image sensor.

17. The camera module of claim 16, comprising: a driving unit disposed on an insulating layer of the substrate for the image sensor and configured to move the image sensor.

18. The camera module of claim 17, wherein the driving unit moves the image sensor in at least one direction among an optical axis direction and a direction perpendicular to the optical axis direction.
